

Discrete time signals and systems

Discrete time signals

- Discrete time: Discrete in time only (sampled but not quantised)

Digital: Discrete in time and amplitude (sampled and quantised)

- Discrete time signals $x = (\dots, x_{-1}, x_0, x_1, \dots)$ typically have real/complex elements, and could be right-sided, left-sided, two-sided or finite-length

↳ Right-sided: starts at x_m and $x_k = 0$ for $k < m$ [start at x_0 : semi-infinite]

↳ Left-sided: ends at x_m and $x_k = 0$ for $k > m$

↳ Two-sided: no highest/lowest index for non-zero values

↳ Finite-length: starts at x_m and ends at x_{m+l-1} (length l).

- A discrete time signal may be a sampled continuous signal, in which case, w/ reference to the original continuous (sample $x_k = x(kT)$), we can denote the discrete time signal as

$$x = (x(0), x(kT), x(2kT), \dots)$$

where T is the sampling interval

- We could treat a discrete time signal as a sequence of no. / has a normalised sampling freq. of 1/T Hz.

Discrete time systems

- A discrete time system takes discrete time signal $x[k]$ and produces discrete time signal $y[k]$.

- The defn of linear time-invariant (LTI) system L extends from continuous to discrete time

If for i/p $\{u[k]\}$ we get o/p $\{y[k]\}$ and for i/p $\{v[k]\}$ we get o/p $\{z[k]\}$, then

↳ Linear:

$$L(\alpha\{u[k]\} + \beta\{v[k]\}) = \alpha L(\{u[k]\}) + \beta L(\{v[k]\}) = \alpha\{y[k]\} + \beta\{z[k]\}$$

↳ Time-invariant:

$$L(\{u[k-n]\}) = \{y[k-n]\}$$

- Discrete LTI systems are governed by difference eqns w/ const. coeffs.

Discrete convolution

- The convolution of the sequences $\{a[k]\}$ and $\{b[k]\}$ is the sequence $\{y[k]\}$.

$$y[k] = \sum_{n=-\infty}^{\infty} a[n]b[k-n]$$

- The summation limits can be simplified for the special cases:

↳ Semi-infinite sequence $\{a[k]\}_{k \geq 0}$ and $\{b[k]\}_{k \geq 0}$

$$y[k] = \sum_{n=0}^k a[n]b[k-n]$$

↳ Finite-length sequence $\{a[k]\}_{k=0}^{L-1}$

$$y[k] = \sum_{n=0}^{\min(L,k)} a[n]b[k-n]$$

Convolution and polynomial multiplication

- We can map semi-infinite sequences to power series, i.e. $\{a_k\} \rightarrow a(D)$

$$a(D) = a_0 + a_1 D + a_2 D^2 + \dots$$

The convolution of semi-infinite sequences is equivalent to multiplication of power series.

$$\begin{aligned} a(D)b(D) &= \left(\sum_{i=0}^{\infty} a_i D^i\right) \left(\sum_{j=0}^{\infty} b_j D^j\right) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} a_i b_j D^{i+j} = \sum_{i=0}^{\infty} \sum_{k=i}^{\infty} a_i b_{k-i} D^k \quad (\text{sub } k=i+j) \\ &= \sum_{k=0}^{\infty} \left(\sum_{i=0}^k a_i b_{k-i}\right) D^k = \sum_{k=0}^{\infty} y_k D^k = y(D) \rightarrow \boxed{a(D)b(D) = y(D)} \end{aligned}$$

where we used $\sum_{i=0}^{\infty} \sum_{k=i}^{\infty} \leftrightarrow \sum_{k=0}^{\infty} \sum_{i=0}^k$

$$\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & (0,0) & (0,1) & (0,2) \\ 1 & & (1,1) & (1,2) \end{array}$$

- D is a dummy variable - we never plan to assign a value to the variable $\rightarrow$ formal power series.

- As we do not assign values to the dummy variable, we do not need to worry about the radius of convergence (sufficient for mapping convolutions to multiplications)

* Z-transform is not a formal power series as we do assign a complex value to the variable z .

Z-transform

Z-transform

- For a semi-infinite sequence $\{a_k\}_{k \geq 0}$, the one-sided Z-transform is defined as

$$\boxed{Z[\{a_k\}] = A(z) = \sum_{k=0}^{\infty} a_k z^{-k}}$$

For a two-sided sequence $\{a_k\}_{k=-\infty}^{\infty}$, the two-sided Z-transform is defined as

$$Z[\{a_k\}] = A(z) = \sum_{k=-\infty}^{\infty} a_k z^{-k}$$

- For the two-sided Z-transform, the sequence $\{a_k\}$ is not uniquely determined by $A(z)$, so we must also specify the radius of convergence (not an issue for one-sided transforms)

Properties of Z-transform

① Linearity: $\boxed{Z[\alpha\{a_k\} + \beta\{b_k\}] = \alpha Z[\{a_k\}] + \beta Z[\{b_k\}]}$, $\boxed{Z\left[\frac{1}{\alpha}\{a_k\}\right] = \frac{1}{\alpha} Z[\{a_k\}]}$

- The Z-transform only involves doing linear operations on the sequence values,

$$Z[\alpha\{a_k\} + \beta\{b_k\}] = \sum_{k=0}^{\infty} (\alpha a_k + \beta b_k) z^{-k} = \alpha \sum_{k=0}^{\infty} a_k z^{-k} + \beta \sum_{k=0}^{\infty} b_k z^{-k} = \alpha Z[\{a_k\}] + \beta Z[\{b_k\}]$$

It also follows that if the sequence defn involves only parameter α , i.e. $a_k = f(\alpha)$,

$$Z\left[\frac{1}{\alpha}\{a_k\}\right] = \sum_{k=0}^{\infty} \left(\frac{1}{\alpha} a_k\right) z^{-k} = \frac{1}{\alpha} \sum_{k=0}^{\infty} a_k z^{-k} = \frac{1}{\alpha} Z[\{a_k\}]$$

② Conjugate symmetry: $\boxed{A(\bar{z}) = \overline{A(z)}}$

- The Z-transform has conjugate symmetry for real-valued signals $\{a_k\}$.

$$A(\bar{z}) = \sum_{k=0}^{\infty} a_k \bar{z}^{-k} = \overline{\left(\sum_{k=0}^{\infty} a_k z^{-k}\right)} = \overline{A(z)}$$

- For the two-sided transform, there is the reverse property that for $\{a_k\}$ s.t. $a_k = \bar{a}_{-k}$, the Z-transform of $\{a_k\}$, $A(z)$ is real on the unit circle.

(e.g. $a_k = a + bj$, $a_{-k} = a - bj$ and $z = e^{j\theta}$, $A(e^{j\theta}) = a_0 + \sum_{k=1}^{\infty} (a + bj)e^{jk\theta} + (a - bj)e^{jk\theta} = a_0 + 2 \sum_{k=1}^{\infty} a \cos(k\theta) + b \sin(k\theta)$)

③ Time delay / time advance: $z[\{a_{k-1}\}] = z^{-1}A(z) + a_{-1}$; $z[\{a_{k+1}\}] = zA(z) - za_0$

- Defining the time delay operation: $\{a_k\} \rightarrow \{a_{k-1}\}$.

$$z[\{a_{k-1}\}] = \sum_{k=-\infty}^{\infty} a_{k-1} z^k = z^{-1} \sum_{k=-\infty}^{\infty} a_{k-1} z^{(k-1)} = z^{-1} \sum_{k=-\infty}^{\infty} a_k z^k + a_{-1} = z^{-1}A(z) + a_{-1}$$

Assuming $a_{-1}=0$, then $z[\{a_{k-1}\}] = z^{-1}A(z)$.

- Defining the time advance operation: $\{a_k\} \rightarrow \{a_{k+1}\}$

$$z[\{a_{k+1}\}] = \sum_{k=-\infty}^{\infty} a_{k+1} z^k = z \sum_{k=-\infty}^{\infty} a_{k+1} z^{k+1} = z \sum_{k=-\infty}^{\infty} a_k z^k - za_0 = zA(z) - za_0$$

+ The time-shift properties can be applied repeatedly for larger shifts.

④ Convolution: $z[\{a_k\} * \{b_k\}] = A(z)B(z)$

- Consider the z-transform of the convolution between $\{a_k\}, \{b_k\}$.

$$z[\{a_k\} * \{b_k\}] = \sum_{k=-\infty}^{\infty} \left(\sum_{i=-\infty}^k a_i b_{k-i} \right) z^k = \sum_{i=-\infty}^{\infty} \sum_{k=i}^{\infty} a_i b_{k-i} z^k = \sum_{i=-\infty}^{\infty} \sum_{j=0}^{\infty} a_i b_j z^{(i+j)} = \left(\sum_{i=-\infty}^{\infty} a_i z^i \right) \left(\sum_{j=-\infty}^{\infty} b_j z^j \right) = A(z)B(z)$$

where we used $\sum_{k=-\infty}^{\infty} \sum_{i=-\infty}^k \leftrightarrow \sum_{i=-\infty}^{\infty} \sum_{k=i}^{\infty}$ as before

Common z-transforms

① Geometric: $z[q^k] = \frac{1}{1-qz^{-1}}$; $z[1] = \frac{1}{1-z^{-1}}$; $z[e^{ak}] = \frac{1}{1-e^{a}z^{-1}}$

- Consider the geometric sequence $a_k = q^k$.

$$z[q^k] = \sum_{k=-\infty}^{\infty} q^k z^k = \sum_{k=-\infty}^{\infty} (qz^{-1})^k = \frac{1}{1-qz^{-1}}$$

$q=1 \rightarrow$ step sequence, $a_k=1$

$$z[1] = \frac{1}{1-z^{-1}}$$

$q=e^a \rightarrow$ exponential sequence, $a_k=e^{ak}$

$$z[e^{ak}] = \frac{1}{1-e^a z^{-1}}$$

② sine/cosine: $z[\sin \omega k] = \frac{z \sin \omega}{1 - 2z^{-1} \cos \omega + z^{-2}}$; $z[\cos \omega k] = \frac{1 - z^{-2} \cos \omega}{1 - 2z^{-1} \cos \omega + z^{-2}}$

- Applying linearity, we can get the z-transform for the sine/cosine sequences.

$$a_k = \sin \omega k: z[\sin \omega k] = z\left[\frac{1}{2j}(e^{j\omega k} - e^{-j\omega k})\right] = \frac{1}{2j} \left(\frac{1}{1 - e^{j\omega} z^{-1}} - \frac{1}{1 - e^{-j\omega} z^{-1}} \right) = \frac{z^{-1} \sin \omega}{1 - 2z^{-1} \cos \omega + z^{-2}}$$

$$a_k = \cos \omega k: z[\cos \omega k] = z\left[\frac{1}{2}(e^{j\omega k} + e^{-j\omega k})\right] = \frac{1}{2} \left(\frac{1}{1 - e^{j\omega} z^{-1}} + \frac{1}{1 - e^{-j\omega} z^{-1}} \right) = \frac{1 - z^{-2} \cos \omega}{1 - 2z^{-1} \cos \omega + z^{-2}}$$

③ scale by r^k : $z[r^k g_k] = G(r^{-1}z)$

- Scaling a sequence $\{g_k\}$ w/ a geometric sequence r^k , i.e. $a_k = r^k g_k$.

$$z[r^k g_k] = \sum_{k=-\infty}^{\infty} r^k g_k z^k = \sum_{k=-\infty}^{\infty} g_k (r^{-1}z)^k = G(r^{-1}z)$$

④ Ramp: $z[k] = \frac{z^{-1}}{(1-z^{-1})^2}$

- Differentiating the geometric sequence,

$$z[kq^k] = z\left[\frac{d}{dq} q^k\right] = \frac{d}{dq} z[q^k] = \frac{d}{dq} \frac{1}{1-qz^{-1}} = \frac{z^{-1}}{(1-qz^{-1})^2}$$

$q=1 \rightarrow$ ramp sequence, $a_k=k$

$$z[k] = \frac{z^{-1}}{(1-z^{-1})^2}$$

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Initial value theorem (IVT) and final value theorem (FVT)

- The IVT states that

$$\lim_{z \rightarrow \infty} A(z) = a_0$$

$$\rightarrow \lim_{z \rightarrow \infty} A(z) = \lim_{z \rightarrow \infty} \left(\sum_{k=0}^{\infty} a_k z^{-k} \right) = \lim_{z \rightarrow \infty} (a_0 + a_1 z^{-1} + a_2 z^{-2} \dots) = a_0$$

- The FVT states if all the poles of $(z-1)A(z)$ lie strictly within the unit circle,

$$\lim_{z \rightarrow 1} (z-1)A(z) = \lim_{k \rightarrow \infty} a_k$$

$\rightarrow$ All poles of $A(z)$ are in the unit circle, except possibly a pole at $z=1$.

Assuming distinct poles, $A(z) = \frac{\alpha_0}{1-z^{-1}} + \sum_i \frac{\alpha_i}{1-p_i z^{-1}}$ where $|p_i| < 1$

By inverse z-transform, $a_k = \alpha_0 + \sum_i \alpha_i p_i^k$

Taking limits $k \rightarrow \infty$, $\lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} (\alpha_0 + \sum_i \alpha_i p_i^k) = \alpha_0$

consider $(z-1)A(z) = z(1-z^{-1})A(z) = z\alpha_0 + (z-1) \sum_i \frac{\alpha_i}{1-p_i z^{-1}}$

Taking limits $z \rightarrow 1$, $\lim_{z \rightarrow 1} (z-1)A(z) = \lim_{z \rightarrow 1} (z\alpha_0 + (z-1) \sum_i \frac{\alpha_i}{1-p_i z^{-1}}) = \alpha_0$

$$\therefore \alpha_0 = \lim_{k \rightarrow \infty} a_k = \lim_{z \rightarrow 1} (z-1)A(z)$$

Z-transforms and other transforms

z-transform and Laplace transform

- For continuous time signals $x(t)$, the Laplace transform is defined as

$$\bar{x}(s) = \int_0^{\infty} x(t) e^{-st} dt$$

Analogously, we can define a transform for a discrete time signal $\{x(kT)\}_{k \geq 0}$, as

$$\sum_{k=0}^{\infty} x(kT) e^{-skT}$$

If we define $x_k = x(kT)$ and $z = e^{sT}$, we get the z-transform

$$X(z) = \sum_{k=0}^{\infty} x_k z^{-k}$$

- Alternatively, we can obtain the z-transform by taking the Laplace transform of a continuous time representation of a discrete time signal, $x^*(t)$.

$$x^*(t) = \sum_{k=0}^{\infty} x(kT) \delta(t - kT)$$

$$\therefore L[x^*(t)] = \int_0^{\infty} \sum_{k=0}^{\infty} x(kT) \delta(t - kT) e^{-st} dt = \sum_{k=0}^{\infty} x(kT) e^{-skT} = \sum_{k=0}^{\infty} x_k z^{-k} = Z[x_k]$$

Z-transform and DTFT

- consider a band limited signal $x(t)$ where $x(t)=0$ for $t < 0$.

The DTFT of the sampled signal is defined as

$$X(\theta) = \sum_{k=-\infty}^{\infty} x(kT) e^{j\theta k}$$

where $\theta = \omega T$ is the normalised frequency.

If we define $x_k = x(kT)$ and $z = e^{j\theta}$, we get the Z-transform

$$X(z) = \sum_{k=-\infty}^{\infty} x_k z^{-k}$$

i.e. $X(z=e^{j\theta}) = X(\theta)$, i.e. Z-transform on the unit circle is the DTFT.

- Since the Z-transform on the unit circle is the DTFT, we can express the IDTFT as

$$x_k = x(kT) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta) e^{j\theta k} d\theta = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) e^{j\theta k} d\theta$$

i.e. we can do an inverse Z-transform using $x_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) e^{j\theta k} d\theta$

- The Z-transform on the unit circle shows the normalised Fourier spectrum

(i.e. normalised frequency $\theta \in [-\pi, \pi]$ and actual frequency $\omega \in [-f_s/2, f_s/2]$).

- For real-valued signals, the conjugate symmetric property states $X(e^{j\theta}) = \overline{X(e^{-j\theta})}$

(i) $|X(e^{j\theta})| = |X(e^{-j\theta})|$ and (ii) $\angle X(e^{j\theta}) = -\angle X(e^{-j\theta})$

Inverse Z-transform

Inverse Z-transform

- The inverse Z-transform can be found using

↳ Partial fractions

↳ Polynomial long division

↳ Taylor expansion.

↳ IDTFT formula.

① Partial fractions

- Decompose the rational function into partial fractions and use standard Z-transforms

- e.g.: $A(z) = \frac{z^3}{z^2(z-1)(z-2)}$ Find $\{a_k\}_{k \geq 0}$.

$$A(z) = \frac{z^3}{z^2(z-1)(z-2)} = z \left[\frac{2}{z-1} - \frac{1}{z-2} \right] = z^3 \left[\frac{1}{z-1} - \frac{1}{z-2} \right] \rightarrow a_k = \begin{cases} 0 & \alpha k \leq 2 \\ 2(1)^{k-3} - 2^{k-3} & k \geq 3 \end{cases}$$

② Polynomial long division

- Explicitly compute the coefficients of $z^0, z^{-1}, \dots \rightarrow$ first few terms of $\{x_k\}$. [Not analytic]

- e.g.: $A(z) = \frac{z^3}{z^2(z-1)(z-2)}$ Find $\{a_k\}_{k \geq 0}$.

$$\begin{aligned} A(z) &= \frac{z^3}{z^2(z-1)(z-2)} = \frac{1}{z} \frac{z^3}{z^2-3z+2} \\ &\quad \begin{array}{r} z^3 + 0z^2 - 3z + 2 \\ \underline{z^3 - 3z + 2} \\ -2z^1 + 6z^0 + 4z^1 \\ \underline{-2z^1 + 6z^0 + 4z^1} \\ -6z^0 + 4z^1 \end{array} \rightarrow A(z) = z^{-2}(z^1 - 2z^0 - 6z^{-1} + \dots) \\ &\quad \begin{array}{r} -2z^1 + 6z^0 + 4z^1 \\ \underline{-6z^0 + 4z^1} \\ -6z^0 + 4z^1 \end{array} = z^{-3}(1 - 2z^{-2} - 6z^{-3} + \dots) \end{aligned}$$

$$\therefore \{a_k\}_{k \geq 0} = \{0, 0, 0, 1, 0, -2, -6, \dots\}$$

③ Taylor expansion

- Use the Taylor expansion to get a power series $\rightarrow$ read off the coefficients.

- eg. $A(z) = \ln(1+z^{-1})$ Find $\{a_k\}_{k \geq 0}$.

$$A(z) = \ln(1+z^{-1}) = z^{-1} - \frac{1}{2}z^{-2} + \frac{1}{3}z^{-3} - \frac{1}{4}z^{-4} + \dots \rightarrow \{a_k\}_{k \geq 0} = \{0, 1, -1/2, 1/3, -1/4, \dots\}$$

④ IDTFT formula.

- Given the Z-transform on the unit circle, we can use $a_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} A(e^{j\theta}) e^{jk\theta} d\theta$.

- eg: $A(z=e^{j\theta}) = \begin{cases} 1 & \text{if } \theta \in [-\pi/4, \pi/4] \\ 0 & \text{o/w} \end{cases}$

$$a_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} A(e^{j\theta}) e^{jk\theta} d\theta = \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} 1 \cdot e^{jk\theta} d\theta = \frac{1}{\pi k} \frac{1}{j} (e^{jk\pi/4} - e^{-jk\pi/4}) = \frac{\sin(k\pi/4)}{\pi k} = \frac{1}{4} \operatorname{sinc}\left(\frac{k\pi}{4}\right).$$

System transfer function

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System transfer function

Pulse response of LTI systems

- We define the unit pulse (Kronecker delta function) δ_k in discrete time as

$$\delta_k = \begin{cases} 1 & k=0 \\ 0 & k \neq 0 \end{cases}$$

- If the i/p to a system is the unit pulse input $\{x_k\}_{k \geq 0} = \delta_k = (1, 0, 0, \dots)$, then the o/p to a system $\{h_k\}_{k \geq 0} = (h_0, h_1, h_2, \dots)$ is the pulse response of the system.

- For a LTI system w/ pulse response $\{h_k\}_{k \geq 0}$, we can write the general i/p $\{x_k\}_{k \geq 0}$ in the form

$$x_k = \sum_{n=0}^k x_n \delta_{k-n}$$

Using the superposition principle, the o/p of the system is thus

$$y_k = \sum_{n=0}^k x_n h_{k-n}$$

Z-transfer function

- The system described by the following difference eqn, subject to zero initial conditions is LTI.

$$y_k + a_1 y_{k-1} + \dots + a_n y_{k-n} = b_0 x_k + b_1 x_{k-1} + \dots + b_m x_{k-m} \quad , \quad x_k = y_k = 0 \text{ for } k < 0$$

Taking the Z-transform,

$$Y(z) [1 + a_1 z^{-1} + \dots + a_n z^{-n}] = X(z) [b_0 + b_1 z^{-1} + \dots + b_m z^{-m}]$$

We define the transfer function $G(z)$ to be

$$G(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_m z^{-m}}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_n z^{-n}}$$

Pulse response and Z-transfer function

- The o/p $\{y_k\}_{k \geq 0}$ of a LTI system can be found by convolving the i/p $\{x_k\}_{k \geq 0}$ w/ the pulse response $\{h_k\}_{k \geq 0}$ of the system

$$y_k = \sum_{n=0}^k x_n h_{k-n} \quad \text{or} \quad \{y_k\} = \{x_k\} * \{h_k\}$$

Taking the Z-transform gives

$$Y(z) = H(z) X(z)$$

where $H(z) = \frac{Y(z)}{X(z)}$ is the system transfer function.

→ The system transfer function $H(z)$ is the Z-transform of the system pulse response $\{h_k\}_{k \geq 0}$.

$$H(z) = Z[\{h_k\}]$$

Solution of linear difference equations

- Take the Z-transform on the linear difference eqn, applying the linearity and time shift property.

- Substitute in any initial conditions and/or inputs

- Solve for $Y(z)$ and take the inverse Z-transform to find $\{y_k\}_{k \geq 0}$.

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Finite impulse response (FIR) and infinite impulse response (IIR)

- FIR filters have pulse responses that terminate after a finite no. of time steps.

$$\{h_k\}_{k \geq 0} = \{h_0, h_1, \dots, h_n, 0, 0, \dots\}$$

IIR filters have pulse responses that do not terminate after a finite no. of time steps

$$\{h_k\}_{k \geq 0} = \{h_0, h_1, h_2, \dots\}$$

- FIR systems have transfer functions of the special form

$$G(z) = b_0 + b_1 z^{-1} + \dots + b_n z^{-n} = \frac{b_0 z^n + b_1 z^{n-1} + \dots + b_n}{z^n}$$

where $a_k = 0$ (no feedback), and all the poles are at $z = 0$.

IIR systems have transfer functions in the general form

$$G(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_n z^{-n}}{1 + a_1 z^{-1} + \dots + a_n z^{-n}}$$

where the poles are at arbitrary locations.

Causality

- Discrete-time systems whose pulse response $\{h_k\}$ where $h_k = 0$ for $k < 0$ are causal.

- consider a non-causal pulse response $\{g_k\} = \{\dots, g_{-1}, g_0, g_1, \dots\}$.

By convolution, the o/p of the system is

$$y_k = \sum_{n=-\infty}^{\infty} g_n x_{k-n} = \dots \underbrace{g_{-2} x_{k+2} + g_{-1} x_{k+1}}_{\text{future}} + \underbrace{g_0 x_k + g_1 x_{k-1} + \dots}_{\text{past}}$$

i.e. the current value of the o/p y_k depends on future values of the i/p $x_{k+1}, x_{k+2}, \dots$

- In causal systems, only the past or an i/p sequence $x_k, x_{k-1}, \dots$ can influence the current o/p y_k .
- The transfer function of causal systems have the form

$$H(z) = h_0 + h_1 z^{-1} + h_2 z^{-2} + \dots$$

For a transfer function w/ the following form to be causal

$$H(z) = \frac{b_0 z^n + b_1 z^{n-1} + \dots + b_n}{z^n + a_1 z^{n-1} + \dots + a_n}$$

we require $n \leq N$ o/w we can rewrite the transfer function as

$$H(z) = c_1 z^{n-n} + \dots + c_{m-1} z + \frac{b_0 z^n + b_1 z^{n-1} + \dots + b_n}{z^n + a_1 z^{n-1} + \dots + a_n}$$

which includes non-causal terms $z, z^2, \dots, z^{m-1}$

+ Note for causal systems, $\lim_{z \rightarrow \infty} G(z) = g_0 < \infty$.

- Systems that operate in function of time and in real time are always causal.
- Examples of non-causal "real world systems" include

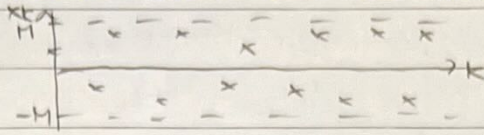
↳ Image processing (function of location)

↳ signal processing system w/ stored/buffered data (not in real time).

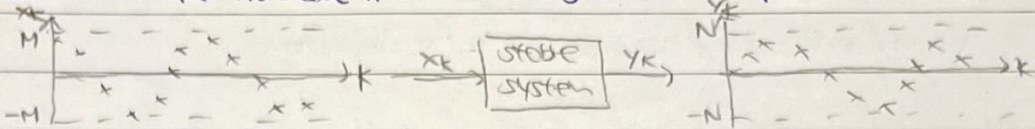
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Stability

- A signal $\{x_k\}$ is bounded if there exists a const. M s.t. $|x_k| < M \forall k$



- A discrete time system is stable if bounded i/p's give bounded o/p's (BIBO stability)



- Let G be a discrete time system w/ a rational transfer function

causal system

$$G(z) = \frac{b(z)}{a(z)} = \frac{b_0 z^m + \dots + b_n}{z^n + a_1 z^{n-1} + \dots + a_n}$$

where $m \leq n$ and there are no common factors b/w $n(z)$ and $d(z)$.

- Let the pulse response of G be $\{g_k\}_{k \geq 0}$, then the following are equivalent: (i) G is stable, (ii) All of the roots of $a(z)$, p_i , satisfy $|p_i| < 1$, and (iii) $\sum_{k=0}^{\infty} |g_k|$ is finite.

↳ Proof of (i) $\rightarrow$ (ii) (Using contrapositive, $\neg(ii) \rightarrow \neg(i)$, i.e. $\neg(|p_i| < 1 \forall i) \rightarrow G$ not stable)

• Suppose $p_1, p_2, \dots$ are distinct. Then we can decompose G using partial fractions

$$G(z) = \frac{\alpha_1}{1 - p_1 z^{-1}} + \dots + \frac{\alpha_n}{1 - p_n z^{-1}} \rightarrow g_k = \alpha_1 p_1^k + \dots + \alpha_n p_n^k$$

• Suppose $|p_i| > 1$ for some i , then g_k is unbounded.

$\therefore$ A pulse i/p (bounded) gives an unbounded o/p, so G is unstable.

• Suppose we have a pole on the unit circle, i.e. $p_i = e^{j\theta}$ for some i . G can be rewritten as

$$G(z) = \frac{\alpha_i}{1 - e^{j\theta} z^{-1}} + \tilde{G}(z) \rightarrow g_k = \alpha_i e^{j\theta k} + \dots$$

The term that corresponds to the pole on the unit circle $g_k = \alpha_i e^{j\theta k} \rightarrow$ bounded

• Now consider the i/p $U(z) = \frac{z^{-1}}{1 - e^{j\theta} z^{-1}}$, which corresponds to the sequence $u_{k-1} \rightarrow$ bounded.

$$\text{The corresponding o/p is } Y(z) = G(z)U(z) = \left(\frac{\alpha_i}{1 - e^{j\theta} z^{-1}} + \tilde{G}(z) \right) \left(\frac{z^{-1}}{1 - e^{j\theta} z^{-1}} \right) = \frac{\alpha_i z^{-1}}{(1 - e^{j\theta} z^{-1})^2} + \tilde{G}(z)U(z)$$

where the first term corresponds to the sequence $y_k = \alpha_i^2 k e^{j\theta(k-1)} \rightarrow$ unbounded

$\therefore$ A bounded i/p gives an unbounded o/p, so G is unstable

$\Rightarrow$ If G is stable, then all of the $a(z)$, p_i , satisfy $|p_i| < 1$.

↳ Proof of (ii) $\rightarrow$ (iii) (i.e. $|p_i| < 1 \forall i \rightarrow \sum_{k=0}^{\infty} |g_k| < \infty$)

• Suppose $p_1, p_2, \dots$ are distinct. Then we can decompose G using partial fractions.

$$G(z) = \frac{\alpha_1}{1 - p_1 z^{-1}} + \dots + \frac{\alpha_n}{1 - p_n z^{-1}} \rightarrow g_k = \alpha_1 p_1^k + \dots + \alpha_n p_n^k$$

$$\therefore \sum_{k=0}^{\infty} |g_k| \leq \alpha_1 \sum_{k=0}^{\infty} |p_1|^k + \dots + \alpha_n \sum_{k=0}^{\infty} |p_n|^k \text{ where RHS is finite if } |p_i| < 1 \forall i$$

↳ Proof of (iii) $\rightarrow$ (i) (i.e. $\sum_{k=0}^{\infty} |g_k| < \infty \rightarrow G$ is stable)

• Let $\{x_k\}$ be an arbitrary bounded i/p, i.e. $|x_k| < M$ for $k \geq 0$. Then the o/p is given by

$$y_k = \sum_{n=0}^{\infty} g_n x_{k-n} \rightarrow |y_k| = \left| \sum_{n=0}^{\infty} g_n x_{k-n} \right| \leq \sum_{n=0}^{\infty} |g_n| |x_{k-n}| < \sum_{n=0}^{\infty} |g_n| M = M \sum_{n=0}^{\infty} |g_n|$$

$\therefore$ Given $\sum_{n=0}^{\infty} |g_n| < \infty$, we have $|y_k| < M \sum_{n=0}^{\infty} |g_n| < \infty$ (bounded o/p).

(We can also prove (i) $\rightarrow$ (iii) using contrapositive, w/ $u_{n+k} = \sin(\theta k) \rightarrow y_n = \sum_{k=0}^{\infty} g_k u_{n+k} = \sum_{k=0}^{\infty} |g_k| > M$)

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Transient and steady state response

Transient and steady state for discrete-time systems.

- The general sol'n of a linear difference eqn. is

$$\{y_k\}_{k \geq 0} = \{t_k\}_{k \geq 0} + \{s_k\}_{k \geq 0}$$

where $\{t_k\}$ is the transient response and $\{s_k\}$ is the steady state response.

- The transient response $\{t_k\}$ decays for stable systems.

Pulse response and poles.

- Consider a LTI system $G(z)$ subjected to a pulse i/p $\{x_k\} = \{1, 0, 0, \dots\}$, so $\Delta(z) = 1$.

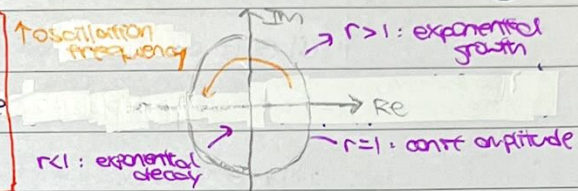
$$\text{The o/p } \{y_k\} \text{ is given by } y_k = \mathcal{Z}^{-1}[Y(z)] = \mathcal{Z}^{-1}[G(z)\Delta(z)] = \mathcal{Z}^{-1}[G(z)]$$

i.e. the poles of $G(z)$ define the response to a pulse.

- In general, for a pole of $z = re^{j\theta}$,

↳ the magnitude r gives the amplitude decay rate

↳ the argument θ gives the oscillation rate



• Recall all the poles inside the unit circle $|p_i| < 1 \forall i \rightarrow$ stable system.

- Consider the following pole locations

↳ Real poles $[p = \lambda] : g_k = \lambda^k$

$$G(z) = \frac{1}{1 - \lambda z^{-1}} \rightarrow g_k = \lambda^k$$

↳ Conjugate symmetric poles $[p_1 = \lambda e^{j\theta}, p_2 = \lambda e^{-j\theta}] : g_k = 2\lambda^k \cos(k\theta)$

$$G(z) = \frac{1}{1 - \lambda e^{j\theta} z^{-1}} + \frac{1}{1 - \lambda e^{-j\theta} z^{-1}} \rightarrow g_k = \lambda^k (e^{j\theta k} + e^{-j\theta k}) = 2\lambda^k \cos(k\theta)$$

• Real-valued impulse responses have complex conjugate pair poles.

↳ Repeated poles $[p_1 = \lambda, p_2 = \lambda] : g_k = (1+k) \lambda^k$

$$G(z) = \frac{1}{(1 - p z^{-1})^2} = \frac{z^{-1}}{(1 - p z^{-1})^2} \rightarrow g_k = (1+k) p^k$$

Stable filters and steady state response.

- In the z-domain, any linear filter can be written as

$$A(z) Y(z) = B(z) X(z) + C(z, y_i)$$

where $C(z, y_i)$ takes into account initial conditions of the system

- Rearranging for $Y(z)$ and taking the inverse z-transform,

$$y_k = \mathcal{Z}^{-1} \left[\frac{B(z)}{A(z)} X(z) + \frac{C(z, y_i)}{A(z)} \right]$$

$$\text{As } k \rightarrow \infty, \quad \lim_{k \rightarrow \infty} y_k = \lim_{k \rightarrow \infty} \mathcal{Z}^{-1} \left[\frac{B(z)}{A(z)} X(z) + \frac{C(z, y_i)}{A(z)} \right] = \lim_{k \rightarrow \infty} \mathcal{Z}^{-1} \left[\frac{B(z)}{A(z)} X(z) \right]$$

- For a stable system, stable roots in $A(z)$ enforce the exponential decay of $\mathcal{Z}^{-1} \left[\frac{C(z, y_i)}{A(z)} \right]$,

whereas $\frac{B(z)}{A(z)} X(z)$ may not necessarily decay exponentially $\rightarrow$ stable filter 'forgets' initial conditions.

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steady state response to a step input

- Provided all the poles of $(z-1)Y(z) = (z-1)G(z)U(z)$ lie strictly within the unit circle, we can use FVT,

$$\lim_{k \rightarrow \infty} y_k = \lim_{z \rightarrow 1} (z-1)G(z)$$

- For a step i/p, $U(z) = \frac{1}{1-z^{-1}}$, so the steady state value of the dp is given by $\lim_{k \rightarrow \infty} y_k = G(1)$

$$\lim_{k \rightarrow \infty} y_k = \lim_{z \rightarrow 1} (z-1)Y(z) = \lim_{z \rightarrow 1} (z-1)G(z)U(z) = \lim_{z \rightarrow 1} (z-1)G(z)\frac{z}{z-1} = \lim_{z \rightarrow 1} zG(z) = G(1)$$

(For the continuous case, FVT: $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sG(s)$, $U(s) = \frac{1}{s}$, so $\lim_{t \rightarrow \infty} y(t) = G(0)$ for step i/p)

steady state response to a sinusoidal input

- Consider the complex sinusoidal i/p $x_k = e^{j\theta k}$ to a system $G(z) = \frac{b(z)}{(z-p_1)(z-p_2)\dots(z-p_n)}$

$$Y(z) = G(z)X(z) = \frac{b(z)}{(z-p_1)(z-p_2)\dots(z-p_n)} \frac{1}{1-e^{j\theta}z^{-1}} = \frac{G(e^{j\theta})}{1-e^{j\theta}z^{-1}} + \text{terms of the form } \frac{B_i}{1-p_i z^{-1}}$$

(In general, $Y(z) = \frac{P_0}{1-e^{j\theta}z^{-1}} + \text{terms of the form } \frac{B_i}{1-p_i z^{-1}}$, consider $[(1-e^{j\theta}z^{-1})Y(z)]|_{z=e^{j\theta}}$

$$[(1-e^{j\theta}z^{-1})Y(z)]|_{z=e^{j\theta}} = P_0, [(1-e^{j\theta}z^{-1})G(z)U(z)]|_{z=e^{j\theta}} = G(e^{j\theta}) \rightarrow P_0 = G(e^{j\theta})$$

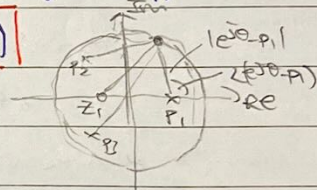
- Taking the inverse z-transform,

$$y_k = G(e^{j\theta})e^{j\theta k} + \text{terms decaying to 0 as } k \rightarrow \infty$$

$$\therefore \text{The stationary dp is } \hat{y}_k = G(e^{j\theta})e^{j\theta k} = |G(e^{j\theta})| e^{j(\theta k + \angle G(e^{j\theta}))}$$

- Now consider the real sinusoidal i/p $x_k = \cos(\theta k) = \frac{1}{2}(e^{j\theta k} + e^{-j\theta k})$. Applying linearity,

$$\hat{y}_k = \frac{|G(e^{j\theta})|}{2} (e^{j(\theta k + \angle G(e^{j\theta}))} + e^{-j(\theta k + \angle G(e^{j\theta}))}) = |G(e^{j\theta})| \cos(\theta k + \angle G(e^{j\theta}))$$



Bode diagrams

- A bode diagram plots the filter amplification $|G(e^{j\theta})|$ and phase shift $\angle G(e^{j\theta})$ at each frequency θ .

- Consider a transfer function of the form $G(z) = C \frac{\prod_{k=1}^m (z-z_k)}{\prod_{k=1}^n (z-p_k)}$

$$\rightarrow |G(e^{j\theta})| = |C| \frac{\prod_{k=1}^m |e^{j\theta} - z_k|}{\prod_{k=1}^n |e^{j\theta} - p_k|} \rightarrow |G(e^{j\theta})|_{dB} = 20 \log(|G(e^{j\theta})|) = 20 \log(|C|) + \sum_{k=1}^m \log |e^{j\theta} - z_k| - \sum_{k=1}^n \log |e^{j\theta} - p_k|$$

i.e., $|G(e^{j\theta})|$ = product of distances from zeroes to $e^{j\theta}$ divided by product of distances from poles to $e^{j\theta}$

(in dB, the products/divisions become sums/subtractions)

$$\rightarrow \angle G(e^{j\theta}) = \angle C + \sum_{k=1}^m \angle(e^{j\theta} - z_k) - \sum_{k=1}^n \angle(e^{j\theta} - p_k)$$

i.e. $\angle G(e^{j\theta})$ = sum of angles from zeroes to $e^{j\theta}$ minus sum of angles from poles to $e^{j\theta}$.

- When $e^{j\theta}$ is close to a pole, the magnitude of the response rises (resonance)

When $e^{j\theta}$ is close to a zero, the magnitude of the response falls (null)

- Since $e^{j\theta k_0} = e^{j(\theta k_0 + 2\pi)}$, going more than a complete revolution is redundant

- For real-valued signals, the conjugate symmetric property of z-transform means $G(p e^{j\theta}) = \overline{G(e^{j\theta})}$,

i.e. $|G(e^{j\theta})| = |G(p e^{j\theta})|$ and $\angle G(e^{j\theta}) = -\angle G(p e^{j\theta})$, so -ve frequencies mirror the frequencies.

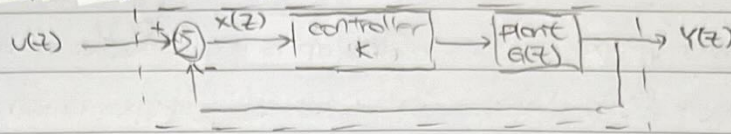
- Therefore, bode diagrams are generally limited to $\theta \in [0, \pi]$ and $\omega \in [0, f_s/2]$.

- The gain and phase margins can be read off the bode diagram as in the continuous time case

Nyquist stability criterion

closed-loop control systems

- Consider the closed-loop control system below.



where we will assume the open loop system is causal (but not necessarily assume it is stable)

$$X(z) = U(z) - Y(z); \quad Y(z) = KG(z)X(z) \rightarrow Y(z) = \frac{U(z) \cdot KG(z)}{1 + KG(z)}$$

$$\therefore H(z) = \frac{Y(z)}{U(z)} = \frac{KG(z)}{1 + KG(z)}$$

* same form as for continuous time signals.

Nyquist diagrams

- The Nyquist diagram of G is $G(e^{j\theta})$ plotted on the complex plane for $\theta \in [-\pi, \pi]$. To draw the Nyquist plot,
 - Read from the Bode diagram, obtain the real/imaginary parts using the gain/phase.
 - substitute various values of θ into $G(e^{j\theta})$ and plot directly (try values like $\theta=0, \theta=\pi/2, \theta=\pi$).
- If there are open-loop poles on the unit circle, the Nyquist trajectory, $G(e^{j\theta})$ will go to infinity as it passes through the pole on the unit circle.
- For real-valued systems, the conjugate symmetry property of z-transforms gives $G(e^{j\theta}) = \overline{G(e^{-j\theta})}$, thus to find a complete Nyquist diagram, we only need to plot $G(e^{j\theta})$ for $0 \leq \theta \leq \pi$, then draw its complex conjugate locus.

Nyquist stability criterion.

- The Nyquist stability criterion allows us to predict the stability of the closed loop $H(z) = \frac{KG(z)}{1 + KG(z)}$ from the Nyquist diagram of the open loop G .
- The Nyquist stability criterion states that the closed loop system is stable iff the no. of encirclements of $-1/K$ by $G(e^{j\theta})$ as θ increases from $-\pi$ to π equals the no. of open loop unstable poles $N_{p,0}$.
- Our goal is to vary K to shift the pt $-1/K$ on the real axis s.t. we get a stable closed loop system.
- 1 The interpretation is the same as the continuous-time Nyquist diagram - gain and phase margins can be read off in the same way and have the same meanings.

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Encirclements of the origin by $F(e^{j\theta})$.

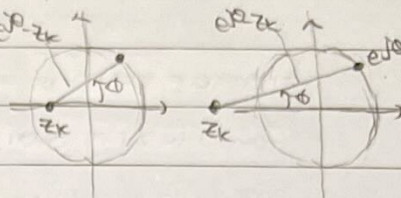
- consider the rational function $F(z) = A \frac{\prod_{k=1}^M (z - z_k)}{\prod_{k=1}^N (z - p_k)}$. The phase of $F(e^{j\theta})$ is given by.

$$\angle F(e^{j\theta}) = \angle A + \sum_{k=1}^M \angle (e^{j\theta} - z_k) - \sum_{k=1}^N \angle (e^{j\theta} - p_k)$$

- As θ increases from 0 to 2π (or equivalently $-\pi$ to π),

(i) $|z_k| < 1$, then the increase in $\angle (e^{j\theta} - z_k)$ is 2π

(ii) $|z_k| > 1$, then the increase in $\angle (e^{j\theta} - p_k)$ is 0.



+ similar for poles but neg effect

- Let $N_{z,i}/N_{p,i}$ be the no. of zeros/poles of $F(z)$ inside the unit circle

- The overall increase α of $\angle F(e^{j\theta})$ as θ increases from 0 to 2π is

$$\alpha = 2\pi (N_{z,i} - N_{p,i})$$

The no. of encirclements C of the origin by the Nyquist trajectory $F(e^{j\theta})$ as θ increases from 0 to 2π is

$$C = N_{z,i} - N_{p,i}$$

Encirclements of the origin by $1 + KG(z)$.

- The open loop transfer function is $H(z) = G(z) = \frac{b(z)}{a(z)}$ → open loop poles are solns to $a(z) = 0$

The closed loop transfer function is $H_c(z) = \frac{KG(z)}{1 + KG(z)}$ → closed loop poles are solns to $1 + KG(z) = \frac{a(z) + Kb(z)}{a(z)} = 0$.

- Consider $1 + KG(z) = \frac{a(z) + Kb(z)}{a(z)}$.

→ The zeros of $1 + KG(z)$ (soln to $a(z) + Kb(z) = 0$) are the closed loop poles

→ The poles of $1 + KG(z)$ (soln to $a(z) = 0$) are the open loop poles.

- The no. of zeros/poles of $1 + KG(z)$ inside the unit circle, $N_{z,i}/N_{p,i}$ is equivalent to the no. of closed/open loop poles inside the unit circle $N_{cp,i}/N_{op,i}$, i.e.

$$N_{z,i} = N_{cp,i}$$

$$N_{p,i} = N_{op,i}$$

- The no. of encirclements C of the origin by the Nyquist trajectory $1 + KG(e^{j\theta})$ as θ increases from 0 to 2π is

$$C = N_{z,i} - N_{p,i} = N_{cp,i} - N_{op,i}$$

Pole arithmetic

- Let m/n be the degree of $b(z)/a(z)$. Since the open-loop system $G(z) = \frac{b(z)}{a(z)}$ is causal, $m \leq n$.

Note that $\deg(a(z) + Kb(z)) = \max(\deg(a(z)), \deg(Kb(z))) = \max(n, m) = n$.

- Therefore, $1 + KG(z) = \frac{a(z) + Kb(z)}{a(z)}$ is a fraction of two polynomials of degree n → both have

n roots → no. of zeros = no. of poles = n .

- Let the subscript cp/op denote closed/open loop poles, z/p denote zeros/poles and i/o denote inside/outside the unit circle. [As before, $N_{cp} = N_z$, $N_{op} = N_p$]

$$n = \overset{N_{cp}}{N_{cp,i}} + \overset{N_{op}}{N_{cp,o}} = \overset{N_{op}}{N_{op,i}} + \overset{N_{op}}{N_{op,o}}$$

∴ The no. of encirclements C satisfies $C = N_{cp,i} - N_{op,i} = (n - N_{cp,o}) - (n - N_{op,o}) = N_{op,o} - N_{cp,o}$

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Encirclements of the point $-1/k$ by $G(e^{j\theta})$.

- When $1 + KG(e^{j\theta})$ encircles the origin, equivalently,

→ $KG(e^{j\theta})$ encircles the pt. -1

→ $G(e^{j\theta})$ encircles the pt. $-1/k$.

- The no. of encirclements C of the pt. $-1/k$ by the Nyquist trajectory $G(e^{j\theta})$ as θ increases from 0 to 2π is

$$C = N_{ap,0} - N_{cp,0}$$

- The closed loop system is stable if $N_{cp,0} = 0$, i.e. no closed loop poles outside the unit circle.

Therefore, for closed loop stability, we req.

$$C = N_{ap,0} = N_{p,0}$$

as stated in the Nyquist stability criterion.

Open loop poles on the unit circle

- The Nyquist trajectory $G(e^{j\theta})$ goes to infinity as it passes an open-loop pole on the unit circle, so

to correctly count the no. of encirclements, we need to obtain a closed curve for the Nyquist locus.

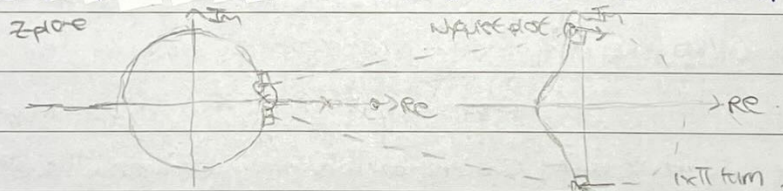
- To close the Nyquist trajectory, we indent the path of z around the poles on the unit circle

w/ a small semi-circular excursion outside the unit circle.

- This means open loop poles on the unit circle are counted as stable in the stability criterion.

i.e. $N_{ap,0} = N_{p,0}$ is the no. of open loop poles strictly outside the unit circle.

- The "right turn" in the z -plane then gives a "right turn" in the Nyquist plot, and a circular arc of large radius is produced which continues for $m\pi$ rad, where m is the multiplicity of the unit-circle pole.



- If $m=1$, the Nyquist locus will be asymptotic to a straight line as it tends to infinity.

→ Suppose that $G(z)$ has a pole at $z=1$, i.e. $G(z) = \frac{F(z)}{z-1}$, where $F(z)$ has no poles/zeros at $z=1$.

Taylor expanding about $z=1$, $G(z) = \frac{F(z)}{z-1} = \frac{1}{z-1} [F(1) + F'(1)(z-1) + \dots] \approx \frac{F(1)}{z-1} + F'(1)$

Note that $\frac{1}{e^{j\theta}-1} = \frac{1}{e^{j\theta/2}(e^{j\theta/2}-e^{-j\theta/2})} = \frac{e^{-j\theta/2}}{e^{j\theta/2}-e^{-j\theta/2}} = \frac{\cos(\theta/2) - js\sin(\theta/2)}{2j\sin(\theta/2)} = -\frac{1}{2} - \frac{j}{2\sin(\theta/2)}$

∴ $G(e^{j\theta}) = F(1) + \frac{F'(1)}{e^{j\theta}-1} = F(1) - \frac{1}{2}F(1) - \frac{jF'(1)}{2\sin(\theta/2)} \rightarrow G(e^{j\theta}) \rightarrow F(1) - \frac{1}{2}F(1)$ as $\theta \rightarrow 0$

→ Consider $G(e^{j\theta})$ directly, then Taylor expand the $e^{j\theta}$ terms, $e^{j\theta} = 1 + j\theta + \frac{(j\theta)^2}{2!} + \dots$

Take out a factor of $j\theta$ from the $(e^{j\theta}-1)$ term, then expand to get the form $\frac{1}{j\theta}(a + bj\theta + \dots)$

The x-coord of the asymptote is b .

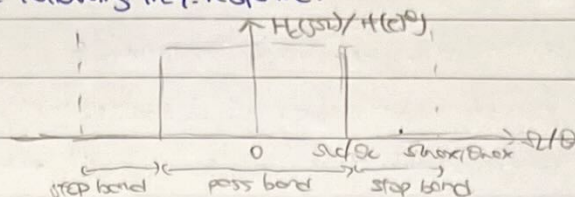
→ May need to use the binomial expansion $(1+k\theta)^{-1} = 1 - k\theta + o(\theta^2)$

Digital filters

Design of filters

Ideal filters

- The ideal LPF has the following freq. response.



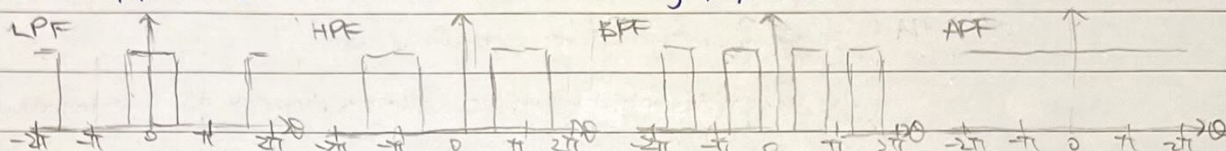
$$\text{Analog: } H_c(j\Omega) = \begin{cases} 1 & \text{if } |\Omega| \leq \Omega_c \\ 0 & \text{if } |\Omega| > \Omega_c \end{cases}$$

$$\text{Digital: } H(e^{j\theta}) = \begin{cases} 1 & \text{if } |\theta| \leq \theta_c \\ 0 & \text{if } |\theta| > \theta_c \end{cases}$$

where $\theta = \omega T$ is the normalised freq. and $T = \frac{1}{f_s} = \frac{2\pi}{\omega_s}$ is the sampling freq.

$$\text{and } \theta_c \leq \theta_{\max} = \pi / \Omega_c \leq \Omega_{\max} = \frac{\omega_s}{2}$$

- Similarly, the ideal HPF/BPF/APF have the following freq. response.



- Consider the ideal LPF. Its impulse response $h[k]$ is found using the DTFT.

$$h[k] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\theta}) e^{j\theta k} d\theta = \frac{1}{2\pi} \int_{-\theta_c}^{\theta_c} e^{j\theta k} d\theta = \frac{1}{2\pi} \frac{e^{j\theta k} - e^{-j\theta k}}{jk} = \frac{\sin(\theta_c k)}{\pi k} = \frac{\theta_c}{\pi} \text{sinc}(\theta_c k)$$

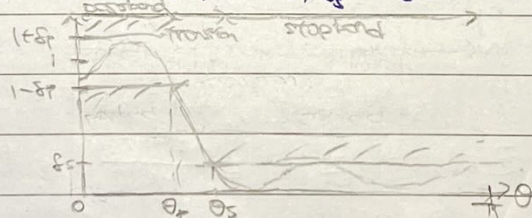
i.e. it is non-causal and has a two-sided infinite impulse response $\rightarrow$ not realisable.

Realisable filter specifications

- A typical filter specification must specify the max. permissible deviations from the ideal, including (i) band edge freq. (θ_p, θ_s) , (ii) max. passband ripple $(1-\delta_p, 1+\delta_p)$, (iii) min. stopband attenuation $(0, \delta_s)$.

- Using the dB scale, we need to specify (i) passband ripple $(20 \log_{10}(1+\delta_p) \text{ dB})$, (ii) peak-to-peak passband ripple $(20 \log_{10}(1+\delta_p) \text{ dB})$, (iii) min. stopband attenuation $(-20 \log_{10}(\delta_s) \text{ dB})$.

($\delta_p = 0.06$ gives) peak-to-peak passband ripple = 2 dB, ($\delta_s = 0.01$ gives) min. stopband attenuation = 40 dB.



FIR filters.

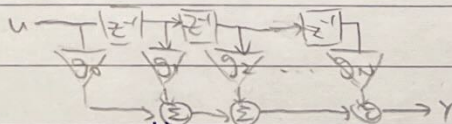
- The FIR filter G is derived by shifting + truncating at $N+1$ samples of the ideal response h , i.e.

$$G(z) = \sum_{k=0}^N g_k z^{-k} \quad \text{where } g_k = h_k - h_{k-N}$$

and h_k is the non-causal impulse response of the ideal filter.

- FIR filters are feed-forward / non-recursive $\rightarrow$ efficiently realised in hardware (FFT).

$$y[n] = \sum_{k=0}^N g_k x[n-k]$$



- FIR filters are inherently stable. $G(z) = \sum_{k=0}^N g_k z^{-k} \rightarrow$ all poles at $z=0$ (inside unit circle)

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IIR filters

- IIR filters have a finite polynomial representation but infinite impulse response.
- IIR filter has feedback — the o/p "goes back" to the filter.

$$y_k = -a_{-1}y_{k-1} + \dots - a_{-N}y_{k-N} + b_0u_k + b_1u_{k-1} + \dots + b_Nu_{k-N}$$

feedback

- IIR filters typically meet a given set of specifications w/ a much lower filter order than their FIR filter counterpart → used for digital implementation of analog filters.

FIR vs IIR filters

- FIR filters:
 - ✓ simple, stable, feed forward, fast implementation (FFT)
 - ✓ can have exactly linear phase $G(e^{j\omega}) = |G(e^{j\omega})| e^{j\theta}$
 - ✓ design methods are generally linear
 - ✗ Higher filter order than IIR filter for same level of performance
 - ✗ often greater delay than IIR filter w/ equal performance.
- IIR filters:
 - ✓ Lower filter order than FIR filter for same level of performance
 - ✓ shorter delay than FIR filter w/ equal performance.
 - ✗ can be unstable, uses feedback.

Design of FIR filters

Frequency distortion of truncation/windowing.

- To get the truncated unit sample response g_k from the shifted desired impulse, we effectively multiply by a rectangular window.

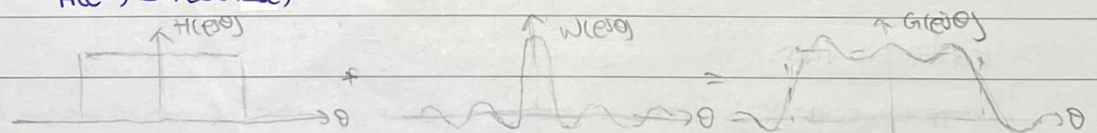
$$g_k = h_k \quad 0 \leq k \leq N \quad \Rightarrow \quad g_k = h_k w_k, \quad w_k = \begin{cases} 1 & 0 \leq k \leq N \\ 0 & \text{o/w} \end{cases}$$

- Using duality of multiplication/convolution for Fourier transforms

$$G(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\sigma}) W(e^{j(\omega-\sigma)}) d\sigma$$

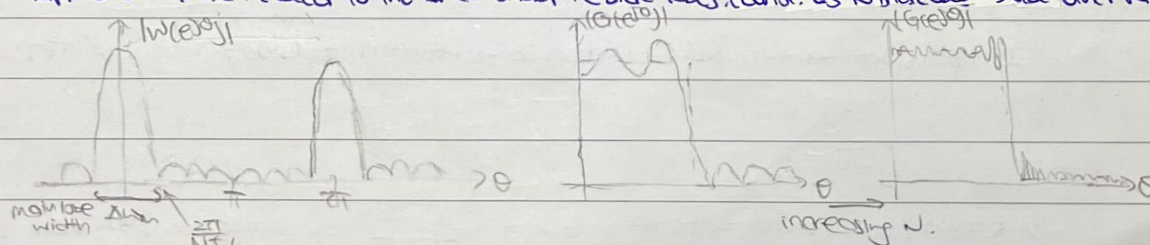
$$\text{where } W(e^{j\omega}) = \sum_{k=0}^N w_k e^{j\omega k} = \sum_{k=0}^N e^{j\omega k} = \frac{(1 - e^{j\omega(N+1)})}{1 - e^{j\omega}} = e^{j\omega \frac{N}{2}} \frac{\sin(\frac{\omega(N+1)}{2})}{\sin(\omega/2)}$$

$$H(e^{j\omega}) = \text{rect}\left(\frac{\omega}{2\omega_c}\right)$$



- The transition band is related to the main lobe, (reduces as $N \rightarrow \infty$)

Ripples of G are related to the area under the side lobes, (const. as N increases → use other windows)



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Design by window method

- We can reduce freq. distortion by adopting diff windows, common windows include:

→ Rectangular	$w_k = \begin{cases} 1 & 0 \leq k \leq N \\ 0 & \text{o/w} \end{cases}$	
→ Triangular (Bartlett)	$w_k = \begin{cases} \frac{2k}{N} & 0 \leq k \leq N/2 \\ 2 - \frac{2k}{N} & N/2 < k \leq N \\ 0 & \text{o/w} \end{cases}$	
→ Hanning	$w_k = \begin{cases} 0.5 - 0.5 \cos(2\pi k/N) & 0 \leq k \leq N \\ 0 & \text{o/w} \end{cases}$	
→ Hamming	$w_k = \begin{cases} 0.54 - 0.46 \cos(2\pi k/N) & 0 \leq k \leq N \\ 0 & \text{o/w} \end{cases}$	

- The shape and duration of the window to control the resulting filter properties:

→ Smaller side lobes yield better approx. of the ideal response

→ Narrow transition bandwidth for increasing N , symmetric at cut-off ω_c .

→ Solve δ for passband error and stopband approx.

	Peak side lobe approx. (relative)	Approx. width of main lobe	Peak approx. error (dB)	
Rectangular	-13	$\frac{4\pi}{N}$	-21	
Bartlett	-25	$\frac{8\pi}{N}$	-25	
Hanning	-31	$\frac{8\pi}{N}$	-44	
Hamming	-41	$\frac{8\pi}{N}$	-53	

- The window method can be used to quickly design filters to approx. a given target response.

- It does not impose amplitude response constraints (passband ripple, stopband attenuation) → need to be used iteratively to produce designs which meet such specs.

- To design a filter using the window method:

→ 1) Select a suitable window function w_k , i.e. Hanning/Bartlett etc.

→ 2) Specify an ideal freq. response H , i.e. LPF/BPF etc.

→ 3) Compute the coeff. for the ideal filter, h_k

→ 4) Delay the ideal filter coeff h_k by $N/2$, $h_{k-N/2}$, then multiply by the window function w_k

→ 5) Evaluate the freq. response of the resulting filter and repeat the above steps iteratively if needed.

Low pass filter design (FIR)

- e.g. Design a LPF w/ passband $\theta_p = 0.2\pi$, stopband $\theta_s = 0.3\pi$ and approx. error $\delta = 0.01$

① Max. allowed approx error in dB = $20 \log_{10} \delta = -40 \text{ dB} > -44 \text{ dB}$ → use Hanning window.

② cut-off freq. $\theta_c = \frac{\theta_p + \theta_s}{2} = 0.25\pi$ main lobe width $\Delta\theta = \theta_s - \theta_p = 0.1\pi = \frac{2\pi}{N} \rightarrow N = 80$

③ Ideal filter $h_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega k} d\omega = \frac{1}{2\pi} \int_{-\theta_c}^{\theta_c} e^{j\omega k} d\omega = \frac{1}{2\pi} \frac{e^{j\omega k}}{jk} \Big|_{-\theta_c}^{\theta_c} = \frac{\sin(\theta_c k)}{\pi k} = \frac{\theta_c}{\pi} \text{sinc}(\theta_c k)$

④ Filter coeff $g_k = h_{k-N/2} w_k = \frac{1}{4} \text{sinc}(\frac{\pi}{4}(k-40)) \frac{1}{2} (1 - \cos(\frac{\pi k}{40})) 1_{k \in [0, 80]}$

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Multi band FIR design.

- We can design HPF/BPF using composition of LAF

$$H_{HPF}[\omega, \pi](e^{j\omega}) = H_{LAF}[\omega, \pi] - H_{LAF}[\omega, \omega] \quad H_{BPF}[\omega, \omega_2](e^{j\omega}) = H_{LAF}[\omega, \omega_2] - H_{LAF}[\omega, \omega_1]$$

where $H_{LAF}[\omega, \omega_c] = \begin{cases} 1 & 0 \leq \omega \leq \omega_c \\ 0 & \text{otherwise} \end{cases}$ and the corresponding $h_{LAF} = \frac{\omega_c}{\pi} \text{sinc}(\omega_c k)$

- The ideal impulse response for HPF/BPF is found by linearity of the transformation,

$$h_{K, HPF} = \text{sinc}(\pi k) - \frac{\omega_c}{\pi} \text{sinc}(\omega_c k) \quad h_{K, BPF} = \frac{\omega_2}{\pi} \text{sinc}(\omega_2 k) - \frac{\omega_1}{\pi} \text{sinc}(\omega_1 k)$$

- The final filter is thus obtained as usual, using,

$$g_k = h_{K, HPF} + h_{K, BPF}$$

Linear phase

- A linear phase filter has freq. response $G(e^{j\omega})$ that satisfies

$$G(e^{j\omega}) = |G(e^{j\omega})| e^{-j\omega \frac{N}{2}}$$

$g_k \rightarrow G(z)$, $g_{k-\frac{N}{2}} \rightarrow z^{-\frac{N}{2}} G(z)$
 $\therefore$ For $z = e^{j\omega}$, we have $G(z) e^{-j\omega \frac{N}{2}}$

i.e. the phase response is a linear function of freq. ($\phi(\omega) = -\omega \frac{N}{2}$).

- For a linear phase filter, all the freq. components of the i/p signal are shifted in time by the same amt $\rightarrow$ no phase distortion in the o/p signal (i/p and o/p signals have to same shape)

- Linear phase is achieved if the filter coeff g_k is symmetric about $N/2$, i.e.

$$g_k = g_{N-k}$$

$$\begin{aligned} G(e^{j\omega}) &= \sum_{k=0}^N g_k e^{j\omega k} = g_0 e^{j\omega 0} + g_1 e^{j\omega 1} + g_2 e^{j\omega 2} + \dots + g_{N-1} e^{j\omega(N-1)} + \dots \\ &= e^{j\omega \frac{N}{2}} (g_0 e^{j\omega \frac{N}{2}} + g_1 e^{j\omega (\frac{N}{2}-1)} + g_2 e^{j\omega (\frac{N}{2}-2)} + \dots + g_{N-1} e^{j\omega (\frac{N}{2}-1)} + \dots) \\ &= e^{j\omega \frac{N}{2}} (2g_0 \cos(\omega \frac{N}{2}) + 2g_1 \cos(\omega (\frac{N}{2}-1)) + \dots) \end{aligned}$$

* Note that the window method gives linear phase filters ($w_k = w_{N-k}$ by defn, so if the desired h_k is symmetric, i.e. $h_k = h_{N-k}$, then $g_k = h_k - w_k$ is also symmetric $\rightarrow$ linear phase).

Design by optimisation.

- Given the ideal filter H and the weighting function w , we can find the optimal filter G (and its coefficients g_k) of length N by optimisation.

- We define a cost/error function $E(\theta)$

$$E(\theta) = w(\theta) [H(\theta) - G(\theta)]$$

- G can be found by minimising

(i) least squares error $\int_{-\pi}^{\pi} E^2(\theta) d\theta$ i.e. $G^*(\theta) = \arg \min_{G(\theta)} \int_{-\pi}^{\pi} E^2(\theta) d\theta$

(ii) max. error $\max_{-\pi \leq \theta \leq \pi} |E(\theta)|$ i.e. $G^*(\theta) = \arg \min_{G(\theta)} \max_{-\pi \leq \theta \leq \pi} |E(\theta)|$

$\rightarrow$ Chebyshev filter.

Design of IIR filters

Discretization by response matching.

- We can convert a continuous time filter $G_c(s)$ into an equivalent discrete time filter $G(z)$ by matching their response to a certain i/p (impulse, step, ramp).
- For a certain i/p, the response is matched if the dp y satisfies

$$L^{-1}[Y(s)]|_{t=kT} = y(kT) = y_k = Z^{-1}[Y(z)]$$

① Impulse invariance

$$G_c(s) \xrightarrow{L^{-1}} g(t) \xrightarrow{\text{sample}} g(kT) = y_k \xrightarrow{Z} G(z)$$

$$\therefore G(z) = Z[L^{-1}[G_c(s)]|_{t=kT}]$$

* For bandlimited filters, the dist'd filter freq. response closely approx to continuous time freq. response

* Impulse invariance preserves stability, i.e. $\text{Re}[p_k] < 0 \rightarrow |e^{p_k T}| < 1$

② Step invariance

$$G_c(s) \xrightarrow{\text{step}} \frac{G_c(s)}{s} \xrightarrow{L^{-1}} y(t) \xrightarrow{\text{sample}} y(kT) = y_k \xrightarrow{Z} Y(z) \xrightarrow{\text{step}} \frac{Z-1}{Z} Y(z)$$

$$\therefore G(z) = \frac{Z-1}{Z} Z[L^{-1}(\frac{G_c(s)}{s})|_{t=kT}]$$

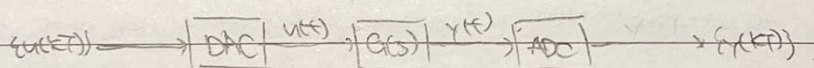
③ Ramp invariance

$$G_c(s) \xrightarrow{\text{ramp}} \frac{G_c(s)}{s^2} \xrightarrow{L^{-1}} y(t) \xrightarrow{\text{sample}} y(kT) = y_k \xrightarrow{Z} Y(z) \xrightarrow{\text{ramp}} \frac{(Z-1)^2}{T^2} Y(z)$$

$$\therefore G(z) = \frac{(Z-1)^2}{T^2} Z[L^{-1}(\frac{G_c(s)}{s^2})|_{t=kT}]$$

choice of waveform for response matching.

- The choice of waveform to achieve response invariance depends on the hybrid analog/digital system.



- Within the hybrid system, it is the properties of the DAC and ADC that influence the choice of waveform for response invariance.

↳ Sample + hold DAC → step invariance.

↳ Moving average DAC → ramp invariance

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Discretization by algebraic transformation

- Algebraic transformations are used to map systems that are not necessarily bandlimited or the cost of some distortion b/w the freq. response of the continuous/discrete time system.

$$H(z) = H_c(s) \mid s = \gamma(z)$$

where $\gamma(z)$ is given by

Forward difference (Euler)

$$s = \frac{z-1}{T}$$

$$\tilde{x} = \frac{x(t+T) - x(t)}{T} \rightarrow sX(s) = \frac{z-1}{T} X(z)$$

Backward difference

$$s = \frac{z-1}{zT}$$

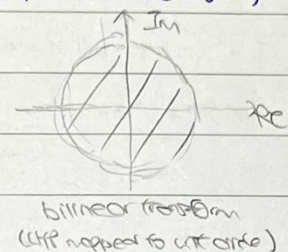
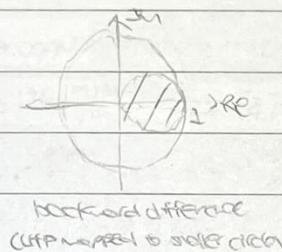
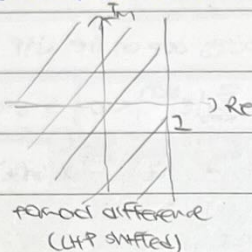
$$\tilde{x} = \frac{x(t) - x(t-T)}{T} \rightarrow sX(s) = \frac{1-z^{-1}}{T} X(z)$$

Bilinear transform (Tustin)

$$s = \frac{z-1}{T} \frac{z+1}{z-1}$$

$$z = e^{sT} = \frac{e^{sT/2}}{e^{-sT/2}} = \frac{1+sT/2}{1-sT/2} \rightarrow s = \frac{2}{T} \frac{z-1}{z+1}$$

- Each of these transformations corresponds to a certain mapping b/w s-plane and z-plane. The shaded region below in the z-plane corresponds to the LHP of the s-plane (stable region)



- Applying algebraic transformation on a stable continuous system results in a stable discrete time system for backward diff / bilinear transform, but is not necessarily true for forward diff.

Bilinear transform

- consider the bilinear transform $s = \gamma(z) = \frac{z-1}{z+1}$

- Analog filter $G_c(s)$ w/ freq. resp. $G_c(j\Omega)$ is Digital filter $G_d(z)$ w/ freq. resp. $G_d(e^{j\theta})$.

Using the algebraic transformation $G_d(z) = G_c(\gamma(z))$, the freq. resp. of the digital filter for $\forall \theta \in \mathbb{R}$ is

$$G_d(e^{j\theta}) = G_c(\gamma(e^{j\theta})) = G_c(j \tan(\theta/2)) = G_c(j\Omega) \quad \text{since } \gamma(e^{j\theta}) = \frac{e^{j\theta}-1}{e^{j\theta}+1} = \frac{e^{j\theta/2}(e^{j\theta/2}-e^{-j\theta/2})}{e^{j\theta/2}(e^{j\theta/2}+e^{-j\theta/2})} = j \tan(\theta/2)$$

- This means the freq. resp. of the analog filter at Ω is mapped to the freq. resp. of the digital filter at θ , where Ω and θ are related by

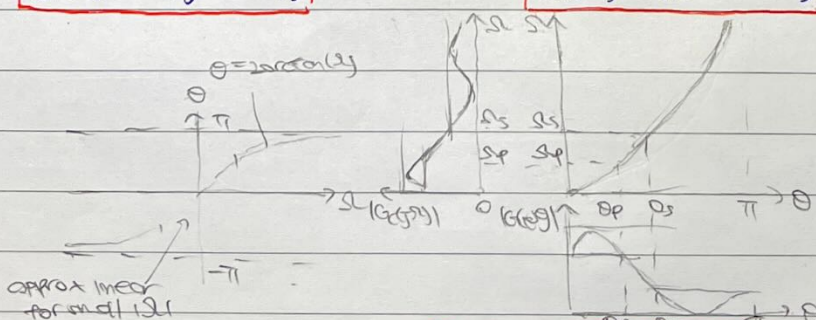
$$\Omega_c = \tan(\theta/2)$$

$$\theta = 2 \arctan(\Omega)$$

- Therefore, we can map b/w the discrete/continuous time filters using

$$G_d(e^{j\theta}) = G_c(j \tan(\theta/2))$$

$$G_c(j\Omega) = G_d(e^{j2 \arctan(\Omega)})$$



- * Using the bilinear transform $s = \gamma(z) = \frac{z-1}{z+1}$, $\Omega_c = \frac{2}{T} \tan(\theta/2)$, $\theta = 2 \arctan(\frac{\Omega T}{2})$, and the mapping is $G_d(e^{j\theta}) = G_c(j \frac{2}{T} \tan(\theta/2))$ $G_c(j\Omega) = G_d(e^{j2 \arctan(\frac{\Omega T}{2})})$

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- consider the bilinear transform $s = \gamma(z) = \frac{z-1}{z+1} \rightarrow z = \gamma^{-1}(s) = \frac{1+s}{1-s}$

$$\text{For } s = \lambda e^{j\omega}, |z|^2 = z z^* = \gamma(s) \gamma^*(s) = \frac{1+\lambda e^{j\omega}}{1-\lambda e^{j\omega}} \cdot \frac{1+\lambda^* e^{-j\omega}}{1-\lambda^* e^{-j\omega}} = \frac{(1+\lambda)^2 + \omega^2}{(1-\lambda)^2 + \omega^2}$$

If $\lambda = 0$, then $|z|^2 = \frac{1+\omega^2}{1+\omega^2} = 1$, i.e. on unit circle

If $\lambda < 0$, then $|z|^2 = \frac{(1+\lambda)^2 + \omega^2}{(1-\lambda)^2 + \omega^2} < 1$, i.e. inside the unit circle.

→ the LHP in the s-plane is mapped exactly to the unit circle in the z-plane → stability preserved.

Low pass filter design (IIR)

- consider the first order analog LPF $G_c(s) = \frac{1}{1+s\tau_c}$, which has gain 1 at $s=j0$ and gain = 0.5 at $s=j\omega_c$. Applying the bilinear transform $s = \gamma(z) = \frac{z-1}{z+1}$,

$$G(z) = G_c(\gamma(z)) = \frac{1}{1+\gamma(z)\tau_c} = \frac{1}{1+\frac{z-1}{z+1}\tau_c} = \frac{(z+1)\tau_c}{(z+1)\tau_c + z-1} = \frac{(z+1)\tau_c}{(1+\tau_c)z + (\tau_c-1)} = \frac{(z+1)\frac{\tau_c}{\tau_c+1}}{z + \frac{\tau_c-1}{\tau_c+1}}$$

- Given a desired cutoff freq. for the digital filter Ω_c , we can find the equivalent pre-warped analog filter cut-off freq. ω_c using $\omega_c = \tan(\Omega_c/2)$.

- e.g. Design a first order digital LPF w/ -3dB freq. of 2kHz and a sampling freq. of 8kHz.

We are given $f_c = 1\text{kHz}$, $f_s = 8\text{kHz} \rightarrow \omega_c = 2\pi f_c$, $T = \frac{1}{f_s} \rightarrow \theta_c = \omega_c T = \frac{2\pi f_c}{f_s} = \frac{\pi}{4}$ rad/sample

$$\therefore \omega_c = \tan\left(\frac{\pi/4}{2}\right) = \tan\left(\frac{\pi}{8}\right) = 0.414 \text{ rad/s} \rightarrow G(z) = \frac{(z+1)\frac{\tau_c}{\tau_c+1}}{z + \frac{\tau_c-1}{\tau_c+1}} = \frac{0.293(z+1)}{z + 0.414}$$

since $G(z) = \frac{Y(z)}{U(z)}$, we have $Y(z)(1 - 0.414z^{-1}) = U(z)(0.293(1+z^{-1}))$, so the difference eq. is

$$y_k = 0.414 y_{k-1} + 0.293 (u_k + u_{k-1})$$

✓ This is an IIR filter as the transfer function $G(z)$ has a pole at $z \neq 0$.

Band transformations

- Analog prototypes are typically LPF. Standard transformations can be used to achieve HPF/BPF/BSF.

- Assuming we have a LPF w/ cutoff at $\omega_c = 1 \text{ rad/s}$.

$$\hookrightarrow \text{LPF} \rightarrow \text{LPF} : s \rightarrow \frac{s}{\omega_c}$$

$$\hookrightarrow \text{LPF} \rightarrow \text{HPF} : s \rightarrow \frac{\omega_c}{s}$$

$$\hookrightarrow \text{LPF} \rightarrow \text{BPF} : s \rightarrow \frac{s^2 + \omega_L^2}{s(\omega_H - \omega_L)}$$

$$\hookrightarrow \text{LPF} \rightarrow \text{BSF} : s \rightarrow \frac{s(\omega_H - \omega_L)}{s^2 + \omega_L \omega_H}$$

} ω_c is the cutoff freq.

} ω_L is the lower cutoff freq.

} ω_H is the upper cutoff freq.

Discrete Fourier Transform

Discrete Fourier Transform (DFT)

- The DFT is given by

$$X_k = \sum_{n=0}^{N-1} x_n e^{-j2\pi kn/N}, \quad \text{for } 0 \leq k \leq N-1$$

- Since the transform maps a finite length sequence $\{x_n\}_{n=0}^{N-1}$ to a finite length sequence $\{X_k\}_{k=0}^{N-1}$, we can rewrite the DFT as a matrix-vector operation.

$$\begin{bmatrix} X_0 \\ X_1 \\ X_2 \\ \vdots \\ X_{N-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & W_N & W_N^2 & \dots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \dots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \dots & W_N^{(N-1)^2} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_{N-1} \end{bmatrix}$$

where $W_N = e^{-j2\pi/N}$ is a primitive N th root of unity (i.e. $W_N^N = 1$ and $W_N^M \neq 1$ for $M=1, 2, \dots, N-1$)

- The N freq. readings obtained from the DFT are "sampled" freq. btwn 0 and 2π in normalised freq. (2π corresponds to sampling freq. f_s).

→ $X_0 \leftrightarrow \text{freq. } 0$; $X_{N/2} \leftrightarrow \text{freq. } \pi$ (i.e. $f_s/2$)

→ $X_1, \dots, X_{N/2-1} \leftrightarrow$ equally spaced pos freq. pts btwn 0 and π (i.e. $f_s/2$)

→ $X_{N/2+1}, \dots, X_{N-1} \leftrightarrow$ equally spaced neg freq. pts btwn 0 and $-\pi$ (i.e. $-f_s/2$)

Cyclic properties of the powers of W_N

- Any integer k can be written as

$$k = qN + r, \quad \text{w/ } 0 \leq r < N$$

where q is the quotient and $r = R_N(k)$ is the remainder

- We can therefore write W_N^k as

$$W_N^k = W_N^{qN+r} = W_N^{qN} \cdot W_N^r = 1 \cdot W_N^r = W_N^r \rightarrow \boxed{W_N^k = W_N^{R_N(k)}}$$

so the DFT Vandermonde matrix $\mathbf{E}$ can be rewritten st. powers of W_N higher than $N/4$ can be expressed as powers btwn 0 and $N/4$.

Constructing the DFT Vandermonde matrix $\mathbf{E}$.

- The DFT Vandermonde matrix $\mathbf{E}$ can be constructed as follows:

$$\mathbf{E} = \begin{bmatrix} 1 & 1 & \dots & \text{(row 0: } N \text{ ones)} \\ 1 & W_N & \dots & \text{(row 1: powers 0 to } N-1 \text{ of } W_N) \\ 1 & W_N^2 & \dots & \text{(row 2: every 2nd element of row 1)} \\ 1 & W_N^3 & \dots & \text{(row 3: every 3rd element of row 1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & \dots & \text{(row } N-1: \text{ row 1 applied } N-1 \text{ backwards)} \end{bmatrix}$$

where "every k th element" is understood cyclically.

→ Note the last row is "every $(N/4)$ th element", which is the same as going backwards.

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Properties of the DFT

① Periodicity: $X_n = X_{n+N}$

$$X_{n+N} = \sum_{k=0}^{N-1} x_k e^{j2\pi k(n+N)/N} = \sum_{k=0}^{N-1} x_k e^{j2\pi kn} e^{j2\pi k} = X_n$$

② Linearity: $\text{DFT}[x_k + y_k] = \text{DFT}[x_k] + \text{DFT}[y_k]$

$$\text{DFT}[x_k + y_k] = \sum_{k=0}^{N-1} (x_k + y_k) e^{j2\pi kn} = \sum_{k=0}^{N-1} x_k e^{j2\pi kn} + \sum_{k=0}^{N-1} y_k e^{j2\pi kn} = \text{DFT}[x_k] + \text{DFT}[y_k]$$

③ Conjugate symmetry: $X_{-n} = X_n^*$ for real x_k ; $x_k = X_{-k}^*$ for real X_n

$$X_{-n} = \sum_{k=0}^{N-1} x_k e^{-j2\pi kn} = \left(\sum_{k=0}^{N-1} x_k e^{j2\pi kn} \right)^* = X_n^*$$

$$x_k = \frac{1}{N} \sum_{n=0}^{N-1} X_n e^{j2\pi kn} = \left(\frac{1}{N} \sum_{n=0}^{N-1} X_n e^{-j2\pi kn} \right)^* = X_{-k}^*$$

i.e. real for time/freq. $\leftrightarrow$ freq/time has cs; complex for time/freq. $\leftrightarrow$ freq/time does not have cs

④ Time/frequency shift: $\text{DFT}[x_{R(k-d)}] = W_N^{-nd} X_n$; $\text{IDFT}[x_{R(n-d)}] = W_N^{kd} x_k$

$$\text{DFT}[x_{R(k-d)}] = \sum_{k=0}^{N-1} x_{R(k-d)} W_N^{kn} = \sum_{k=0}^{N-1} x_k W_N^{nR(k-d)} = \sum_{k=0}^{N-1} x_k W_N^{n(k-d)} = W_N^{-nd} \sum_{k=0}^{N-1} x_k W_N^{kn} = W_N^{-nd} X_n$$

$$\text{IDFT}[x_{R(n-d)}] = \frac{1}{N} \sum_{n=0}^{N-1} x_{R(n-d)} W_N^{-kn} = \frac{1}{N} \sum_{n=0}^{N-1} x_n W_N^{-kR(n-d)} = \frac{1}{N} \sum_{n=0}^{N-1} x_n W_N^{-k(n-d)} = W_N^{kd} \frac{1}{N} \sum_{n=0}^{N-1} x_n W_N^{-kn} = W_N^{kd} x_k$$

⑤ Cyclic convolution: $\text{DFT}[x_k \otimes y_k] = X_n Y_n$ [$\otimes$ is $\otimes$], $R_k = \sum_{m=0}^{N-1} x_m y_{R(k-m)}$

$$\begin{aligned} \text{DFT}[x_k \otimes y_k] &= \sum_{k=0}^{N-1} (x_k \otimes y_k) W_N^{kn} = \sum_{k=0}^{N-1} \left(\sum_{m=0}^{N-1} x_m y_{R(k-m)} \right) W_N^{kn} = \sum_{m=0}^{N-1} x_m \sum_{k=0}^{N-1} y_{R(k-m)} W_N^{kn} \\ &= \sum_{m=0}^{N-1} x_m \sum_{k=0}^{N-1} y_k W_N^{nR(k-m)} = \sum_{m=0}^{N-1} x_m \sum_{k=0}^{N-1} y_k W_N^{n(k-m)} = \sum_{m=0}^{N-1} x_m W_N^{-nm} \sum_{k=0}^{N-1} y_k W_N^{kn} = X_n Y_n \end{aligned}$$

i.e. circular convolution in time domain is equivalent to pt-wise multiplication in freq. domain.

The DFT Vandermonde matrix $\mathbb{E}$.

- The sum of column n of the DFT Vandermonde matrix $\mathbb{E}$ is given by $\sum_{k=0}^{N-1} W_N^{kn}$

$$n=0: \sum_{k=0}^{N-1} W_N^{k \cdot 0} = \sum_{k=0}^{N-1} 1 = N$$

$$n \neq 0: \sum_{k=0}^{N-1} W_N^{kn} = \frac{1 - (W_N^n)^N}{1 - W_N^n} = \frac{1 - W_N^{nN}}{1 - W_N^n} = \frac{1 - 1}{1 - W_N^n} = 0$$

i.e. the sum of any column of $\mathbb{E}$ except column 0 is zero.

- Let $\underline{f}_l$ be the l th column of the DFT Vandermonde matrix $\mathbb{E}$. For $l \geq n$,

$$\text{complex dot product} \rightarrow \underline{f}_l \cdot \underline{f}_n = \sum_{k=0}^{N-1} W_N^{kl} (W_N^{kn})^* = \sum_{k=0}^{N-1} W_N^{k(l-n)} = \sum_{k=0}^{N-1} 1 = N$$

i.e. the sum of the $(l-n)$ th column of $\mathbb{E}$, and hence 0 when $l \neq n$ / N when $l=n$

- This means the columns of $\mathbb{E}$ are orthogonal, and their magnitude is $\sqrt{N}$.

$$\underline{f}_i \cdot \underline{f}_j = \begin{cases} N & i=j \\ 0 & i \neq j \end{cases}$$

$\therefore$ The DFT is an orthogonal basis transform (w/ scaling factor N).

- If $\underline{x} = \mathbb{E}\underline{z}$ and $\underline{y} = \mathbb{E}\underline{z}$, the scaling factor of $\sqrt{N}$ on the basis vectors give

$$\underline{x} \cdot \underline{y} = \left(\frac{1}{\sqrt{N}} \underline{x} \right) \cdot \left(\frac{1}{\sqrt{N}} \underline{y} \right) = \frac{1}{N} \underline{x} \cdot \underline{y}$$

$$\|\underline{x}\|^2 = \left(\frac{1}{\sqrt{N}} \underline{x} \right) \cdot \left(\frac{1}{\sqrt{N}} \underline{x} \right) = \frac{1}{N} \|\underline{x}\|^2$$

INVERSE DISCRETE FOURIER TRANSFORM (IDFT)

- the IDFT is given by

$$x_k = \frac{1}{N} \sum_{n=0}^{N-1} X_n e^{j \frac{2\pi}{N} kn} \quad \text{for } 0 \leq k \leq N-1$$

- since the transform maps a finite length sequence $\{x_n\}_{n=0}^{N-1}$ to a finite length sequence $\{X_k\}_{k=0}^{N-1}$,

we can rewrite the IDFT as a matrix-vector operation,

$$\begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_{N-1} \end{bmatrix} = \frac{1}{N} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & W_N^{-1} & W_N^{-2} & \dots & W_N^{-(N-1)} \\ 1 & W_N^2 & W_N^4 & \dots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \dots & W_N^{(N-1)^2} \end{bmatrix} \begin{bmatrix} X_0 \\ X_1 \\ X_2 \\ \vdots \\ X_{N-1} \end{bmatrix}$$

where $W_N = e^{j \frac{2\pi}{N}}$ is a primitive N th root of unity (i.e. $W_N^N = 1$ and $W_N^m \neq 1$ for $m = 1, 2, \dots, N-1$)

- by inspection, the IDFT is simply DFT but $W_N \rightarrow \frac{1}{N} W_N^{-1}$

Alternatively, we can obtain the IDFT by i/p conjugation $\rightarrow$ DFT $\rightarrow$ o/p conjugation $\rightarrow$ divide by N .

$$\frac{1}{N} [\text{DFT}[X_n^*]]^* = \frac{1}{N} \left[\sum_{n=0}^{N-1} X_n^* e^{j \frac{2\pi}{N} kn} \right]^* = \frac{1}{N} \sum_{n=0}^{N-1} X_n e^{-j \frac{2\pi}{N} kn} = \text{IDFT}[X_n]$$

- Considering the IDFT matrix, it is simply the inverse of the DFT Vandermonde matrix $\underline{E}$, $\underline{E}^{-1}$.

since $\underline{E}$ is orthogonal, it is easy to determine its inverse.

$$\underline{E} \underline{E}^H = N \underline{I}_N \quad \rightarrow \quad \underline{E}^{-1} = \frac{1}{N} \underline{E}^H$$

Note that $\underline{E}$ is symmetric, so we just need to conjugate the entries of $\underline{E}$, i.e. $\underline{E}^{-1} = \frac{1}{N} \underline{E}^*$

- As $W_N^H = (W_N^*)^T$, the inverse matrix $\underline{E} = \underline{E}^*$ has the same row 0 as $\underline{E}$ and its rows 1 to $N-1$ are the corresponding rows of $\underline{E}$ in reverse order (counting down from $N-1$ to 1).

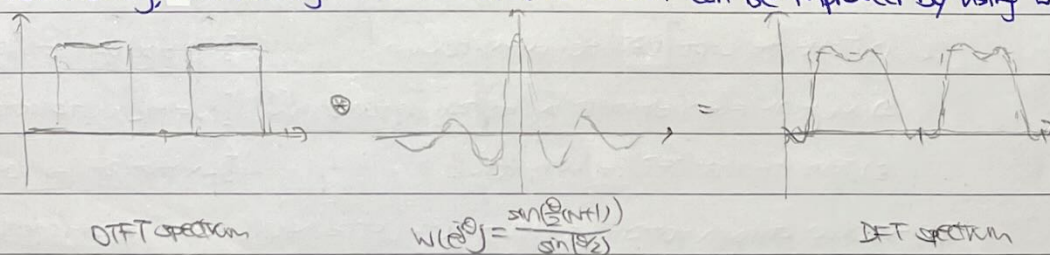
DFT and DTFT frequency comparison.

- the DFT can be interpreted as an approx. of the continuous DTFT spectrum by truncation + sampling.

- Truncation of the sequence $\{x_n\}_{n=0}^{N-1}$ (DTFT $\rightarrow$ DFT) is equivalent to a multiplication of the sequence by a rectangular window of length N . / convolution of the spectrum w/ $W(e^{j\Omega}) = \frac{\sin(\frac{\Omega}{2}(N+1))}{\sin(\frac{\Omega}{2})}$.

- Therefore the value of x_n depends on the whole DTFT spectrum by convolution (instead of only over the ideal interval $[\frac{2\pi}{N}(n-\frac{1}{2}), \frac{2\pi}{N}(n+\frac{1}{2})]$) $\rightarrow$ leakage from neighbouring freq. bins.

- As for filtering, the matching of DFT to DTFT spectrum can be improved by using windowing.



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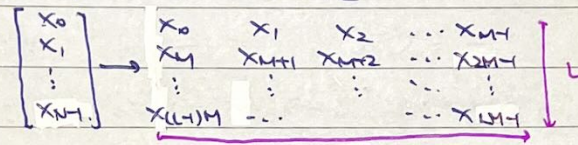
Fast Fourier Transform

Fast Fourier Transform (FFT)

- FFT algorithms exploit properties of the DFT Vandermonde matrix E to perform the DFT in $O(N \log N)$ operations instead of $O(N^2)$ normally needed for matrix-vector multiplication.
- ↳ FFT relies on redundancy in the calculation of the basic DFT → reuse past computations.
- ↳ FFT is recursive → logarithmic complexity.
- There are many FFT algorithms that work for length N w/ specific properties.
- ↳ (General) Cooley Tukey: $N = LM$, where $L, M \in \mathbb{Z}^+$.
- ↳ Radix-2: $N = 2^M$, where $M \in \mathbb{Z}^+$.

Cooley Tukey algorithm.

- The Cooley Tukey algorithm works for composite numbers $N = LM$, where $L, M \in \mathbb{Z}^+$.
- If further factorisation of L/M is possible, the method can be applied recursively.
- Reconsider the N -term i/p and o/p vectors $\underline{x}, \underline{X}$ as a $L \times M$ array w/ L rows and M columns, i.e.



This is row major form — consecutive entries in a row are consecutive in $\underline{x}$; consecutive entries in a column are M apart in $\underline{x}$.

- Reconsidering the time/freq. domain index k/n using a 2D index, we have

$$k = LM + s$$

$$n = ML + r$$

where l, r indicate the row no.; m, s indicate the column no.

- Using the 2D index, we can rewrite the DFT as.

$$X_n = X_{nL+r} = \sum_{k=0}^{N-1} x_k W_N^{kn} = \sum_{s=0}^{M-1} \sum_{l=0}^{L-1} x_{LM+l+s} W_N^{(LM+l+s)(ML+r)} = \sum_{s=0}^{M-1} \sum_{l=0}^{L-1} x_{LM+l+s} W_N^{LMs} W_N^{lM} W_N^{LMr} W_N^{lsr}$$

Noting that $W_N^{LMs} = e^{-j\frac{2\pi}{N}LMs} = 1$, $W_N^L = e^{-j\frac{2\pi}{N}L} = e^{-j\frac{2\pi}{M}} = W_M$, $W_N^M = e^{-j\frac{2\pi}{N}M} = e^{-j\frac{2\pi}{L}} = W_L$,

$$X_{nL+r} = \sum_{s=0}^{M-1} W_M^{sm} W_N^{sr} \sum_{l=0}^{L-1} x_{LM+l+s} W_L^{lr}$$

$\sum_{l=0}^{L-1} x_{LM+l+s} W_L^{lr}$

↑
Twiddle factor L-pe DFT
M-pe DFT.

- 1) Take the L-pe DFT for every column [$M \times L$ -pe DFT → ML^2 computations]
- 2) Multiply every element by its twiddle factor W_N^{sr} [$N = ML$ multiplications → N computations]
- 3) Take the M-pe DFT of every row. [$L \times M$ -pe DFT → LM^2 computations]

- The total no. of computations is $ML + ML^2 + LM^2 = ML(1 + L + M) \approx N(L + M)$

The total no. of computations for the N -pe DFT is N^2 .

Radix-2 algorithm

- The Radix-2 algorithm works for powers of 2, $N=2^M$, where $M \in \mathbb{Z}^+$.

- We can split the summation of the N -pt DFT into two parts.

$$X_n = \sum_{k=0}^{N-1} x_k e^{-j\frac{2\pi}{N}nk} = \sum_{k=0}^{N/2-1} x_{2k} e^{-j\frac{2\pi}{N}(2k)n} + \sum_{k=0}^{N/2-1} x_{2k+1} e^{-j\frac{2\pi}{N}(2k+1)n} = \sum_{k=0}^{N/2-1} x_{2k} e^{-j\frac{2\pi}{N/2}nk} + e^{-j\frac{2\pi}{N}n} \sum_{k=0}^{N/2-1} x_{2k+1} e^{-j\frac{2\pi}{N/2}nk}$$

$\therefore X_n = A_n + W_N^n B_n$, where A_n, B_n are $\frac{N}{2}$ -pt DFTs.

DFT of even index DFT of odd index

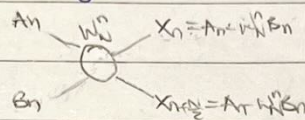
- consider $X_{n+N/2}$

$$X_{n+N/2} = \sum_{k=0}^{N/2-1} x_{2k} e^{-j\frac{2\pi}{N}(n+N/2)k} + e^{-j\frac{2\pi}{N}(n+N/2)} \sum_{k=0}^{N/2-1} x_{2k+1} e^{-j\frac{2\pi}{N}(n+N/2)k} = \sum_{k=0}^{N/2-1} x_{2k} e^{-j\frac{2\pi}{N/2}nk} - e^{-j\frac{2\pi}{N}n} \sum_{k=0}^{N/2-1} x_{2k+1} e^{-j\frac{2\pi}{N/2}nk}$$

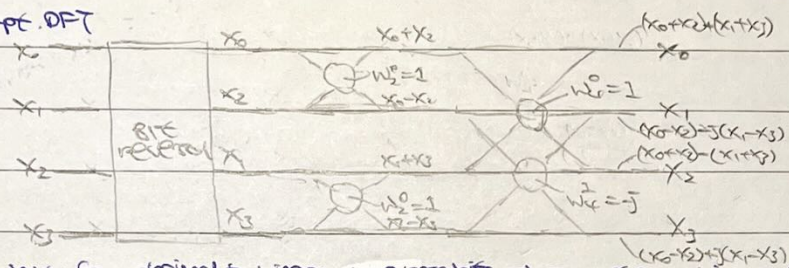
$\therefore X_{n+N/2} = A_n - W_N^n B_n$, where A_n, B_n are $\frac{N}{2}$ -pt DFTs.

DFT of even index DFT of odd index

- We have recursion + redundancy by decomposing a N -pt DFT into $2 \times \frac{N}{2}$ -pt DFTs + twiddle factor computations.



- eg: consider the 4-pt DFT



bit reversal: convert index from decimal to binary $\rightarrow$ reverse bits $\rightarrow$ convert from binary to decimal.

Applications of the DFT

Cyclic convolution

- consider independent RVs $X_1, X_2, \dots, X_n$ defined over an alphabet $\mathcal{X} = \{0, 1, \dots, p-1\}$, where $p \in \mathbb{Z}^+$.
consider the RV Z

$$Z = X_1 \oplus X_2 \oplus \dots \oplus X_n, \text{ where } \oplus \text{ is mod-}p \text{ addition } (x \oplus y = P_p(x+y))$$

For all n and p , the pmf of Z , P_Z , is given by the cyclic convolution of the pmfs of X_i , P_{X_i} .

$$P_Z = P_{X_1} \oplus P_{X_2} \oplus \dots \oplus P_{X_n}$$

- For $p=2$, $X_1, X_2, \dots, X_n$ are binary RVs w/ pmf $P_{X_i} = [1-p_i, p_i]$

$$1) \text{ 2-pt DFT of } P_{X_i}: Q_{X_i} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1-p_i \\ p_i \end{bmatrix} = \begin{bmatrix} 1 \\ 1-2p_i \end{bmatrix}$$

$$2) \text{ component-wise multiplication: } Q_Z = \begin{bmatrix} 1 & 1 \\ 1 & 1-2p_i \end{bmatrix} = \begin{bmatrix} 1 \\ 1-2p_i \end{bmatrix}$$

$$3) \text{ Inverse 2-pt DFT: } P_Z = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1-2p_i \end{bmatrix} = \begin{bmatrix} 1/2 + 1/2 p_i \\ 1/2 - 1/2 p_i \end{bmatrix}$$

- In general, we evaluate the circular convolution using FFT $\rightarrow$ component-wise multiplication $\rightarrow$ IFFT.

OFDM (Frequency = convolution channel in doing cyclic convolution)

(Refer to JEEF notes)

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Linear convolution

- Linear convolutions of arbitrary sequences cannot in general be computed via an FFT.

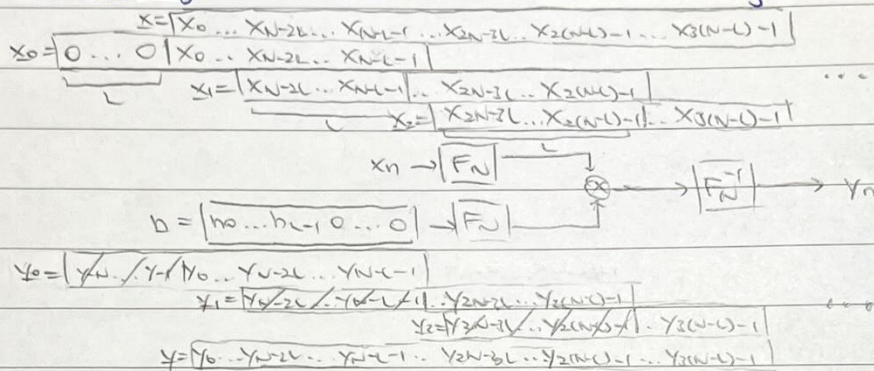
However, consider the case of an FIR filter of length $L+1$.

$$\{y_k\}_{k \geq 0} = \{h_k\}_{k \geq 0} * \{x_k\}_{k \geq 0} \quad \text{re. } y_k = \sum_{m=0}^L h_m x_{k-m}$$

- If we take a block $\underline{x} = [x_0 \dots x_{N-1}]$ and evaluate the circular convolution $y_c = \underline{h} \circledast \underline{x}$, the first L d/s differ from linear convolution because they depend on the last L entries of the ip block, instead of $x_L \dots x_{N-1} = 0$ for linear convolution.

- For $N > L$, the d/s beyond the first L will be identical to linear convolution.

- We can use length N FFT to do linear convolutions w/ FIR of length $L+1 \ll N$ w/ overlap-save method.



Interpolation and decimation.

- signal processing systems often req. a change of sampling freq. f_s so processing can be done at a higher f_s (better accuracy) or lower f_s (reduces computations)

- Interpolation (upping to a higher freq.) is done by first zero stuffing: insert L zeros b/w every two samples, then filtering: low-pass and scale the resulting signal.

- Decimation (reducing to a lower freq.) is done by dropping samples: keep only one in every K samples. (to avoid aliasing, this can only be done when $BW < \frac{f_s^{new}}{2}$)

- Interpolating / decimating via the FFT gives a graceful transition b/w freq. and no longer req. f_s^{new} to be a multiple or divisor of f_s^{old} .

- To interpolate/decimate, we pick N and N' s.t. $N'/N = f_s^{new}/f_s^{old}$.

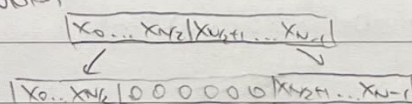
1) Take the N -pt FFT of the i/p $\underline{x}$.

2) Interpolation: Insert zeros b/w $X_{N/2+1}$ and $X_{N/2+1} \pmod{N}$ (odd N) / either side of $X_{N/2}$ (even N)

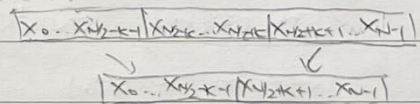
Decimation: Delete components symmetrically around $X_{N/2}$.

3) Take the N' -pt IFFT of the modified spectrum

interpolation



decimation



* The FFT method has the usual caveat of a cut in the spectrum being equivalent to multiplying by a rectangular window $\rightarrow$ can be improved using window-based approaches.

Continuous time random signals

Random process

- A random signal is a time-dependent RV, $X(t)$, which is an instance drawn randomly from a set of possible signal waveforms (ensemble)
- The ensemble of random signals + the associated probability distributions is known as a random process.
- We denote the random process as $\{X(t, \alpha)\}$ and an instance of the random process, the random signal as $x(t, \alpha)$, where α is a draw from some set A .

Correlation and covariance functions

- The autocorrelation function of a random process $\{X(t, \alpha)\}$ is

$$r_{xx}[t_1, t_2] = E_A[X(t_1, \alpha)X(t_2, \alpha)] = \int_A x(t_1, \alpha)x(t_2, \alpha)f_A(\alpha) d\alpha$$

The autocovariance function of a random process $\{X(t, \alpha)\}$ is

$$c_{xx}[t_1, t_2] = E_A[(X(t_1, \alpha) - M_{X,t_1})(X(t_2, \alpha) - M_{X,t_2})] = \int_A (x(t_1, \alpha) - M_{X,t_1})(x(t_2, \alpha) - M_{X,t_2})f_A(\alpha) d\alpha$$

- The crosscorrelation function b/w random processes $\{X(t, \alpha)\}$ and $\{Y(t, \alpha)\}$ is

$$r_{xy}[t_1, t_2] = E_A[X(t_1, \alpha)Y(t_2, \alpha)] = \int_A x(t_1, \alpha)y(t_2, \alpha)f_A(\alpha) d\alpha$$

The crosscovariance function b/w random processes $\{X(t, \alpha)\}$ and $\{Y(t, \alpha)\}$ is

$$c_{xy}[t_1, t_2] = E_A[(X(t_1, \alpha) - M_{X,t_1})(Y(t_2, \alpha) - M_{Y,t_2})] = \int_A (x(t_1, \alpha) - M_{X,t_1})(y(t_2, \alpha) - M_{Y,t_2})f_A(\alpha) d\alpha$$

* Note the expectations $E_A[\cdot]$ are over the whole ensemble.

Stationarity

- stationarity means the statistical characteristics of a signal do not change over time.
- A random process is strict sense stationary (SSS) if its probability distribution does not change over time. Formally, for all finite N and all sets of time pts $\{t_1, \dots, t_N\}$, the N th order joint pdf is invariant for any time shift τ .

$$f_{X(t_1) \dots X(t_N)}(x_1, \dots, x_N) = f_{X(t_1 + \tau) \dots X(t_N + \tau)}(x_1, \dots, x_N)$$

- A random process is wide sense stationary (WSS) iff

$$(i) E[X(t)] = \mu \quad \forall t \text{ (const. mean)}$$

$$(ii) \text{Var}[X(t)] < \infty / E[X(t)^2] < \infty \quad \forall t \text{ (finite variance/power) [ignored in 3F1]}$$

$$(iii) r_{xx}[t_1, t_2] \rightarrow r_{xx}[t_2 - t_1] \quad (r_{xx} = r_{xx}(t_2 - t_1))$$

- For WSS $\{X(t, \alpha)\}$, $r_{xx}[\tau] = r_{xx}[-\tau]$, i.e. $r_{xx}[\tau]$ is even

$$r_{xx}[\tau] = E[X(t)X(t+\tau)] = E[X(t+\tau)X(t)] = r_{xx}[-\tau]$$

Moreover, as τ becomes large, $X(t)$ and $X(t+\tau)$ tends to become decorrelated. ($r_{xx}[\tau] \rightarrow E[X(t)]E[X(t+\tau)] = \mu^2$)

Therefore, $\max |r_{xx}[\tau]| = r_{xx}[0]$ (Formal proof in 3F3 notes)

The width of the autocorrelation function $r_{xx}[\tau]$ tells us how slowly the signal fluctuates in time (width $\propto \frac{1}{\text{freq}}$)

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Power spectral density (PSD)

- For WSS $\{x(t)\}$ the PSD function is the FT of the autocorrelation function $r_{xx}(\tau)$

$$S_x(\omega) = \int_{-\infty}^{\infty} r_{xx}(\tau) e^{-j\omega\tau} d\tau$$

$$r_{xx}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_x(\omega) e^{j\omega\tau} d\omega$$

- For jointly WSS $\{x(t), y(t)\}$, the cross PSD function is the FT of the crosscorrelation function $r_{xy}(\tau)$

$$S_{xy}(\omega) = \int_{-\infty}^{\infty} r_{xy}(\tau) e^{-j\omega\tau} d\tau$$

$$r_{xy}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xy}(\omega) e^{j\omega\tau} d\omega$$

Ergodicity

- If we can exchange ensemble averages $E_A[\cdot]$ for time averages $\langle \cdot \rangle_T$, the random process is ergodic.

- This means the ensemble of the random process simply composed of all possible time shifts of a single random signal.

- Ensemble averages $E_A[\cdot]$ and time averages $\langle \cdot \rangle_T$ are defined as

$$E_A[X(t, \alpha)] = \int_A x(t, \alpha) f_A(\alpha) d\alpha \quad \langle X(t, \alpha) \rangle_T = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t, \alpha) dt$$

- A WSS $\{x(t, \alpha)\}$ could be mean ergodic/correlation ergodic, defined as

↳ mean ergodic

$$\mu = E_A[X(t, \alpha)] = \langle X(t, \alpha) \rangle_T$$

↳ correlation ergodic

$$r_{xx}(\tau) = E_A[x(t, \alpha)x(t+\tau, \alpha)] = \langle x(t, \alpha)x(t+\tau, \alpha) \rangle_T$$

white noise

- In reality, no process is truly stationary (therefore cannot be ergodic), but many noise processes are approx stationary $\rightarrow$ use ergodicity to simplify measurement of WSS process.

- We may estimate quantities from a finite period $[T_1, T_2]$ of a ergodic process using

$$\mu \approx \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} X(t, \alpha) dt$$

$$r_{xx}(\tau) \approx \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} X(t, \alpha) X(t+\tau, \alpha) dt$$

System response to random signals

Linear systems and random processes

- If the i/p $\{x(t)\}$ of a LTI system w/ impulse resp $h(t)$ is WSS, then the o/p $\{y(t)\}$ is also WSS.

(i) Consider the mean $E[y(t)]$

$$E[y(t)] = E\left[\int h(k) x(t-k) dk\right] = \int h(k) E[x(t-k)] dk = \mu_x \int h(k) dk \rightarrow \text{const. over all } t$$

(ii) consider the autocorrelation $r_{xx}(\tau)$

$$\begin{aligned} r_{xx}(\tau) &= E\left[\int h(k_1) x(t-k_1) dk_1 \int h(k_2) x(t+\tau-k_2) dk_2\right] = E\left[\int \int h(k_1) h(k_2) x(t-k_1) x(t+\tau-k_2) dk_1 dk_2\right] \\ &= \int \int h(k_1) h(k_2) E[x(t-k_1) x(t+\tau-k_2)] dk_1 dk_2 = \int \int h(k_1) h(k_2) r_{xx}[\tau + k_1 - k_2] dk_1 dk_2 \\ &= \int h(k_1) [h(\tau + k_1) + r_{xx}[\tau + k_1]] dk_1 = h(\tau) + h(\tau) + r_{xx}[\tau] \rightarrow \text{only a function of } \tau \end{aligned}$$

- The i/p $\{x(t)\}$ and o/p $\{y(t)\}$ of the LTI system is jointly WSS. We already know $\{x(t)\}$, $\{y(t)\}$ are WSS.

$$r_{xy}(\tau) = E[x(t)y(t+\tau)] = E\left[x(t) \int h(k) y(t+\tau-k) dk\right] = \int h(k) E[x(t)y(t+\tau-k)] dk = \int h(k) r_{xy}[\tau-k] dk = h(\tau) + r_{xx}[\tau]$$

- Notice that the correlation functions can be expressed as convolutions

$$r_{xy}(\tau) = h(\tau) + h(\tau) + r_{xx}(\tau)$$

$$r_{xy}(\tau) = h(\tau) + r_{xx}(\tau)$$

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- Taking the FT of the correlation functions expressed as convolutions,

$$S_Y(\omega) = H(\omega) H^*(\omega) S_X(\omega) = |H(\omega)|^2 S_X(\omega)$$

$$S_{XY}(\omega) = H(\omega) S_X(\omega)$$

rearranging for $H(\omega)$,

only magnitude $|H(\omega)| = \sqrt{\frac{S_Y(\omega)}{S_X(\omega)}}$

both magnitude and phase $H(\omega) = \frac{S_{XY}(\omega)}{S_X(\omega)}$

- If an important system is subject to measurable random fluctuations at the i/p, we can measure the system freq. response w/o taking the plant offline. (estimate $r_{xx}(\tau)$, $r_{xy}(\tau)$, $r_{yx}(\tau)$ w/ time averages $\rightarrow S_{xx}(\omega)$, $S_{xy}(\omega)$, $S_{yx}(\omega) \rightarrow H(\omega)$)

Physical interpretation of PSD.

- For a deterministic signal $x(t)$, the instantaneous power is $x(t)^2$ and the average power is $\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t)^2 dt$

- For a random process $\{X(t)\}$, the expected instantaneous power is $E[X(t)^2]$ and the expected average power is $E\left[\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t)^2 dt\right] = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T E[X(t)^2] dt$

- If $\{X(t)\}$ is WSS, then $E[X(t)^2] = r_{xx}[0]$ for all t , so the expected average power is $r_{xx}[0]$.

$$r_{xx}[0] = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega) e^{j\omega 0} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega) d\omega$$

i.e. the area under $S_{xx}(\omega)$ gives the total power of $\{X(t)\}$.

- To find the power of a WSS process $\{X(t)\}$ in a particular freq. band, say $[w_1, w_2]$ s.t. $0 < w_1 < w_2$, consider passing $\{X(t)\}$ through an ideal band-pass filter w/ TF $H(\omega) = \begin{cases} 1 & \text{if } w_1 \leq \omega \leq w_2 \\ 0 & \text{o.w.} \end{cases}$, then find the o/p power $r_{yy}[0]$.

$$r_{yy}[0] = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{yy}(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} |H(\omega)|^2 S_{xx}(\omega) d\omega = \frac{1}{2\pi} \left[\int_{w_1}^{w_2} S_{xx}(\omega) d\omega + \int_{w_1}^{w_2} S_{xx}(\omega) d\omega \right] = \frac{1}{\pi} \int_{w_1}^{w_2} S_{xx}(\omega) d\omega$$

$\rightarrow$ The power of $\{X(t)\}$ in a particular freq. band $w_1 \leq \omega \leq w_2$ is $\frac{1}{\pi} \int_{w_1}^{w_2} S_{xx}(\omega) d\omega$.

White noise process.

- A WSS process $\{X(t)\}$ is termed white noise if

$$C_{XX}(\tau) = \sigma^2 \delta(\tau)$$

where $\delta(\tau)$ is the delta function.

- The PSD $S_X(\omega)$ for white noise is given by

$$S_X(\omega) = \int_{-\infty}^{\infty} r_{XX}(\tau) e^{-j\omega\tau} d\tau = \int_{-\infty}^{\infty} (\sigma^2 \delta(\tau) + M^2) e^{-j\omega\tau} d\tau = \int_{-\infty}^{\infty} \sigma^2 \delta(\tau) e^{-j\omega\tau} d\tau + M^2 \int_{-\infty}^{\infty} e^{-j\omega\tau} d\tau = \sigma^2 + M^2 2\pi \delta(\omega)$$

For zero-mean white noise, $M=0$, so $S_X(\omega) = \sigma^2$, i.e. flat across all freq.

- Note that power for white noise is $E[X(t)^2] = r_{xx}[0] = \sigma^2 + M^2 2\pi \delta(0) \rightarrow \infty$, so pure white noise is unrealistic in practice. (However, we can have "nearly white" processes).

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From discrete time to continuous time

- consider a discrete time x_k that is zero for $k \notin [-\frac{N}{2}+1, \frac{N}{2}]$ and each x_k is sampled independently from a distribution w/ $\mu=0, \sigma^2=1$.

- consider $|x(e^{j\theta})|^2 = x(e^{j\theta}) x^*(e^{j\theta})$

$$|x(e^{j\theta})|^2 = \left(\sum_k x_k e^{-j\theta k} \right) \left(\sum_l x_l e^{j\theta l} \right) = \sum_k w_k^2 + \sum_{k \neq l} w_k w_l e^{j\theta(l-k)}$$

Taking expectations, noting that $E[x_k^2] = 1$ and $E[x_k x_l] = E[x_k] E[x_l] = 0$ for $k \neq l$,

$$E[|x(e^{j\theta})|^2] = N$$

$$\therefore \frac{1}{N} E[|Y(e^{j\theta})|^2] = \frac{1}{N} E[|H(e^{j\theta})|^2 |x(e^{j\theta})|^2] = \frac{1}{N} |H(e^{j\theta})|^2 E[|x(e^{j\theta})|^2] = |H(e^{j\theta})|^2$$

- We can derive the continuous case by taking the limit as sampling time tends to zero.

- consider a continuous time $x_h(t)$ b/w $-T/2$ and $T/2$, given by

$$x_h(t) = \sum_{k=-N/2}^{N/2-1} p(t-kh) w_k$$

where $h = T/N$ is the sampling time, and each w_k is sampled independently from a distribution w/ $\mu=0, \sigma^2 = \frac{1}{h}$.

$p(t) = H(t) - H(t-h)$ is the pulse function.

- The FT of $x_h(t)$, $X_h(j\omega)$ is given by

$$X_h(j\omega) = \text{FT} \left[\sum_{k=-N/2}^{N/2-1} p(t-kh) w_k \right] = \sum_{k=-N/2}^{N/2-1} \text{FT}[p(t-kh)] w_k = \sum_{k=-N/2}^{N/2-1} \left(\frac{1}{j\omega} - \frac{1}{j\omega} e^{-j\omega kh} \right) e^{-j\omega kh} w_k$$

$\text{FT}[H(t)]$ $\text{FT}[H(t-h)]$ time shift by kh

consider $|X_h(j\omega)|^2 = X_h(j\omega) X_h^*(j\omega)$

$$|X_h(j\omega)|^2 = \left(\sum_k w_k \left(\frac{1-e^{-j\omega kh}}{j\omega} \right) e^{-j\omega kh} \right) \left(\sum_l w_l^* \left(\frac{1-e^{j\omega lh}}{-j\omega} \right) e^{j\omega lh} \right) = \sum_k w_k^2 \left(\frac{1-e^{-j\omega kh}}{j\omega} \right) \left(\frac{1-e^{j\omega kh}}{-j\omega} \right) + \sum_{k \neq l} w_k w_l^* \left(\frac{1-e^{-j\omega kh}}{j\omega} \right) \left(\frac{1-e^{j\omega lh}}{-j\omega} \right) e^{j\omega(l-k)h}$$

Taking expectations, noting that $E[w_k^2] = \frac{1}{h}$ and $E[w_k w_l] = E[w_k] E[w_l] = 0$ for $k \neq l$, $h = T/N \rightarrow T = Nh$

$$E[|X_h(j\omega)|^2] = \sum_k E[w_k^2] \left(\frac{(1-e^{-j\omega kh})(1-e^{j\omega kh})}{\omega^2} \right) = \sum_k \left(\frac{1}{h} \right) \left(\frac{(1-\cos(\omega kh))}{\omega^2} \right) = \sum_k h = T$$

$\therefore$ Define $x(t) = \lim_{h \rightarrow 0} x_h(t)$, i.e. white noise, then we have

$$\frac{1}{T} E[|Y(j\omega)|^2] = \frac{1}{T} E[|H(j\omega)|^2 |x(j\omega)|^2] = \frac{1}{T} |H(j\omega)|^2 E[|x(j\omega)|^2] = |H(j\omega)|^2$$